

Wisdom or Whims? Investor Opinions, Trading Strategies, and Market Outcomes*

Shuaiyu Chen[†]

Lin Peng[‡]

Dexin Zhou[§]

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Abstract

We study how retail investors form opinions, using large language models to classify nearly 100 million timestamped messages from a major investing social-media platform. Strategy adoption is highly dynamic, shifting between fundamental (FA) and technical analysis (TA) in response to news, past performance, and social feedback. Message informativeness varies sharply across strategies: FA sentiment positively predicts returns and identifies the informed component of retail order flow, whereas TA sentiment negatively predicts returns and captures the behavioral component. TA sentiment also predicts intense Robinhood buying and short squeezes followed by sharp return reversals, with both patterns amplified by social engagement.

Keywords: Retail trading, Technical analysis, Fundamental analysis, Social finance, Social media, Herding, Short squeeze

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[†]chens@arden.virginia.edu, Darden School of Business, University of Virginia.

[‡]lin.peng@baruch.cuny.edu., Zicklin School of Business, Baruch College.

[§]dexin.zhou@baruch.cuny.edu., Zicklin School of Business, Baruch College.

1. Introduction

Retail trading in U.S. equity markets has expanded dramatically over the past decade: by 2021, retail volume rivaled mutual funds and hedge funds combined (Martin and Wigglesworth, 2021), a post-COVID surge that has stayed historically elevated through 2025.¹ This expansion coincided with the rise of investor-focused social media platforms, which have become central venues where retail investors form and exchange views, as well as with speculative meme stock episodes that prompted regulatory scrutiny of retail brokerages, social media, and market makers (U.S. House Committee on Financial Services, 2021).²

Against this backdrop, retail investors have attracted attention in recent studies (e.g., Boehmer et al., 2021; Barber et al., 2022; Welch, 2022; Eaton et al., 2022; Kogan et al., 2024), generating insights into their role in financial markets. However, how retail investors form their opinions, what strategies they use, how these opinions change over time, and how they link to market outcomes remain insufficiently understood.

We open this “black box” using large language models to analyze a large corpus of investor communications: 96 million messages posted by 840,000 users on StockTwits, covering roughly 7,800 stocks from January 2010 to June 2023. These time-stamped messages allow us to track investor opinions at high frequency and link them directly to price and trading dynamics.

From each message, we extract two attributes: the investment strategy referenced and the sentiment expressed. To classify strategies, we develop and validate a cost-effective, two-stage LLM approach focused on two prominent types: fundamental analysis (FA) and technical analysis (TA). Among all messages, 31% reference an investment strategy; FA and TA account for 46% and 28% of strategy-related messages, respectively.³ Three findings emerge. First, retail

¹According to the *Financial Stability Review* (European Central Bank, November 2025), the share of retail investors in U.S. equity trading volume rose sharply, from an average of 14% during 2010–2019 to an average of 19% since 2020, reaching approximately 21% as of that date (Chart 2.7b). The same report documents a marked rise in the share of zero-days-to-expiration (ODTE) contracts in total S&P 500 options volume, with retail investors accounting for 50%–60% of that activity, further underscoring the growing role of retail participation in market dynamics.

²38% of U.S. adults report regularly obtaining news from Facebook and 35% from YouTube, followed by Instagram and TikTok (20% each), X (12%), and Reddit (9%) (Pew Research Center, 2025).

³There is a long literature documenting the effectiveness of fundamental and technical analyses. For fundamental analysis, see, e.g., Porta et al. (1997) and Abarbanell and Bushee (1997). For technical analysis, see Brown and Jennings (1989), Jegadeesh (1991), Brock et al. (1992), Jegadeesh and Titman (1993), Blume et al. (1994),

investor strategy adoption is highly dynamic and varies with news, past strategy performance (*performance feedback*), and peer engagement (*social feedback*). Second, FA sentiment positively predicts returns while TA sentiment negatively predicts. Third, higher TA sentiment predicts intense buying by Robinhood investors and short squeezes followed by sharp price reversals, with social feedback amplifying both the return and trading patterns. Higher FA sentiment, in contrast, is not associated with similar trading frenzies. The evidence suggests that social media discourse can help detect retail investor decision-making and speak to the theories of experience-based learning and social transmission.

We first establish that strategy use is highly dynamic. While [Cookson and Niessner \(2020\)](#) find that heterogeneity in self-declared investment styles on social media explains trading volume, we find that such static user characteristics explain only about 3% of the variation in strategy choice.⁴ Strategy choice is strongly associated with firm news releases. Around earnings announcements, investors are 33% more likely to discuss FA and 23% less likely to discuss TA, indicating that earnings-relevant news shifts attention from TA toward FA.

Beyond news, strategy choice responds to past performance. When a strategy delivers poor returns, the same investor tends to move away from it in subsequent posts. The economic magnitudes are on the order of 3% to 5% of the mean strategy likelihood. Similar patterns hold for FA. The evidence is consistent with experience-based learning and strategy switching ([Malmendier and Nagel, 2011](#); [Barberis and Jin, 2023](#)). The response is also asymmetric: shifts away from underperforming strategies are stronger than shifts toward outperforming ones, paralleling the asymmetric experience effects in retail selling ([Yeung et al., 2025](#)).

Strategy adoption is also shaped by social feedback. Investors stick with strategies whose prior posts received many likes and shift away from those that did not; the effect amounts to 6% to 8% of the mean strategy likelihood. Social feedback is more strongly associated with switching than is realized performance: an investor’s decision to abandon a TA strategy is 95% more sensitive to receiving few likes than to the strategy’s poor performance. This suggests

[Lo et al. \(2000\)](#), [George and Hwang \(2004\)](#), [Han et al. \(2013\)](#), [Han et al. \(2016\)](#), [Jiang et al. \(2023\)](#), and [Murray et al. \(2024\)](#)). Whether retail investors are able to apply these strategies optimally to achieve similar outcomes is a separate and open question, and one our paper speaks to.

⁴Figure 1 presents an example illustrating this within-investor variation. The two messages are from the same investor, a self-declared technical strategist, yet they employ different approaches: one discusses earnings and valuations (FA), while the other relies on support levels and volume patterns (TA).

that social transmission can powerfully reinforce and propagate certain investment strategies, consistent with [Shiller \(2017\)](#) and [Hirshleifer \(2020\)](#).

Having mapped how investors choose strategies, we turn to whether the expressed opinions are informative, linking post sentiment to future returns. If posts contain useful information about fundamentals, sentiment should positively forecast returns. If posts are pure noise, sentiment should have no predictive power. If posts reflect behavioral biases and price dislocations, sentiment should negatively predict returns as prices correct.

For each stock, we construct daily sentiment measures for TA and FA strategies by aggregating sentiments across messages in each category. Stocks with more positive FA sentiment tend to earn *higher* next-week returns, whereas stocks with more positive TA sentiment earn *lower* returns. Long-short portfolio formed on these sentiment measures yield annualized DGTW-adjusted abnormal returns ([Daniel et al., 1997](#)) of 7.75% and −10.10%, respectively. These patterns indicate that FA discussions reflect information not yet incorporated into prices, whereas TA posts reflect behavioral biases linked to mispricing. Social feedback amplifies the contrast: when posts receive many likes, higher FA sentiment predicts even higher returns, while higher TA sentiment predicts even lower returns.

The TA-versus-FA contrast carries over to retail trading. Two settings where coordinated retail demand is particularly pronounced are especially informative: intense buying by Robinhood investors and short-squeeze episodes. Robinhood users tend to be less experienced investors and are therefore more susceptible to social media influences, making this platform a natural laboratory for our purposes. Social media also played a central role in the short-squeeze episodes during the meme-stock mania of 2020–2021.

For the Robinhood analysis, we use user-holding data to identify intense correlated buying episodes. Higher TA sentiment strongly predicts such episodes, especially on days when TA messages receive many likes. These episodes tend to be followed by sharper return reversals. In contrast, more likes to FA messages dampen the link between FA sentiment and intense buying episodes.

Short squeezes provide a more dramatic test of crowded retail demand. Following [Schultz \(2024\)](#), we identify squeezes as days that combine price runs with substantial short cover-

ing. Higher TA sentiment, when popular, predicts subsequent short squeezes. Post-squeeze reversals are also amplified by higher TA sentiment: the typical squeeze is followed by a -6% next-day return, with higher TA sentiment widening this reversal by roughly an additional 40% . No analogous pattern emerges for FA sentiment.

Beyond these two settings, aggregate retail order flow offers a broader test. Both TA and FA sentiments are strongly associated with contemporaneous retail order flow, but decomposing that flow into strategy-linked components reveals sharp differences in return predictability: the FA-linked component positively predicts subsequent returns, whereas the TA-linked component negatively predicts. Retail order flow is thus not monolithic: the portion linked to FA discussions appears informed, while the portion linked to TA commentary appears behavioral.

Across these tests, FA sentiment is consistent with a wisdom-of-crowds effect, whereby social media discussions anchored in asset fundamentals aggregate valuable information. TA sentiment, by contrast, is consistent with social transmission amplifying behavioral biases: socially propagated, unanchored signals generate crowded trading and price dislocations ([Stein, 2009](#); [Pedersen, 2022](#); [Han et al., 2022](#); [Bastianello and Fontanier, 2025](#)). That the TA-versus-FA contrast maps onto returns, retail trading, and short squeezes, events detrimental even to sophisticated investors like hedge funds, suggests that observing investor opinions on social media helps explain the dynamics of prices and trading dynamics.

We conduct a battery of additional tests and robustness checks. The main results are robust to alternative sentiment measures and Fama-MacBeth estimation in place of pooled panel specifications. They are also stronger in the post-2018 period, when retail participation and social media use expanded sharply. The negative return predictability of TA sentiment weakens significantly when a higher fraction of messages comes from self-reported professionals, suggesting that professional participation tempers speculative trading. Messages classified under “Other Strategies” (OS), covering event-driven, options-related, and other strategy types, exhibit return-predictability and order-flow patterns that mirror TA.

Our evidence speaks to the debate over whether retail trading reflects informed demand or liquidity provision (e.g., [Kaniel et al., 2008](#); [Kelley and Tetlock, 2013](#); [Barrot et al., 2016](#); [Boehmer et al., 2021](#); [Welch, 2022](#)) or noise trading rooted in behavioral biases (e.g., [Barber](#)

and Odean, 2000; Kumar and Lee, 2006; Barber and Odean, 2008; Foucault et al., 2011; Barber et al., 2022; Bryzgalova et al., 2023; Barber et al., 2023b).⁵ Our evidence suggests that both views capture part of the picture: the FA-linked component of retail order flow appears informed, while the TA-linked component is consistent with behavioral biases.

Beyond this retail-trading debate, the paper contributes to the literature on how investors form and transmit beliefs. Recent studies elicit beliefs through surveys (e.g., Choi and Robertson, 2020; Giglio et al., 2021; Chinco et al., 2022; Liu et al., 2022; Meeuwis et al., 2022; Jiang et al., 2024), which deliver clean responses but are limited in scale and frequency.⁶ Observing opinion formation on social media complements this work by tracking belief updating at high frequency across hundreds of thousands of investors, in line with Campbell and Ramadorai (2026)’s call for using alternative data to study investor behavior. Where Cookson et al. (2024) find that average social-media sentiment positively predicts returns, we find that this relation depends crucially on the underlying strategy.

On the transmission side, Cookson et al. (2023) and Han et al. (2023) document echo-chamber effects and sentiment contagion among message consumers. We find that the same social-feedback channel operates on message *senders* as well: investors stick with strategies that earned many likes. Peer approval therefore shapes not only what investors choose to hear but also what they choose to say. This sender channel sharpens the social-media squeeze mechanism of Allen et al. (2025): not all retail chatter triggers squeezes; only the component grounded in technical signals and amplified by community engagement does so.

2. Data and Measurement

Our sample includes common stocks (CRSP share codes 10, 11, and 12) traded on the NYSE, AMEX, and NASDAQ from January 2010 through June 2023. We obtain investor social media data from StockTwits, stock market data from CRSP, accounting data from Compustat,

⁵Barber et al. (2023b) note that retail buying is highly concentrated in trades that perform poorly, whereas order-imbalance studies evaluate hypothetical equal-weighted portfolio strategies rather than the actual returns retail investors earn on trades.

⁶A nascent literature uses large language models to extract the mental models of market participants from analyst reports (Bastianello et al., 2024; Ke, 2024), corporate disclosures and managerial forecasts (Jha et al., 2024, 2025), and fund managers’ policy beliefs (Gao et al., 2025).

retail market order data from TAQ, Robinhood data from RobinTrack, and financial news data from RavenPack.

2.1. *StockTwits Data*

StockTwits is a leading social-media platform for retail investors, where users share opinions about stocks, ETFs, and cryptocurrencies. The platform focuses on financial markets, with “cashtags” (e.g., \$TSLA) used to tag specific ticker symbols.⁷ Through the StockTwits API, we collect 169,509,106 messages posted by 978,071 users on 15,232 tickers between January 2010 and June 2023. Each message comes with a timestamp, the textual content, and, when available, a user-provided sentiment label (“bullish” or “bearish”). At the user level, we observe self-reported investment approach (technical, momentum, fundamental, value, growth, or global macro), investment horizon (day trader, swing trader, position trader, or long-term investor), and trading experience (novice, intermediate, or professional). We apply filters to focus on human-generated content about publicly traded firms: (i) we retain only messages tagging exactly one ticker mapped to a stock in the CRSP common-stock universe; (ii) we drop messages from any user who ever posted more than 1,000 messages in a single day, since such activity is typically algorithmic; (iii) we drop messages originating from third-party platforms, which mostly redistribute press releases or financial news; and (iv) we require both the user identifier and the username to be non-missing. The final sample comprises 96,095,345 messages from 840,846 unique users on 7,834 unique stocks.

2.2. *Measuring Strategy and Sentiment*

We extract two attributes from each StockTwits message: the investment strategy referenced and the directional sentiment (bullish or bearish). Section 2.2.1 describes the LLM-based procedure for classify strategies at the message level. Section 2.2.2 describes the procedure for measuring strategy-specific sentiment.

⁷See [Cookson and Niessner \(2020\)](#) and [Cookson et al. \(2024\)](#) for more details about the StockTwits data.

2.2.1. Classifying Investment Strategies

StockTwits messages are short, filled with abbreviations and colloquial expressions, and often contain non-standard spelling.⁸ These features make it difficult to identify investment strategies using traditional dictionary-based methods.

We address this challenge with a two-step LLM classification procedure. In the first step, we apply GPT-4 Turbo to a small random sample of messages and obtain a probability score for whether each message discusses a given strategy. In the second step, we use these outputs to fine-tune a smaller, more efficient BERT model and apply it to the full dataset. This knowledge-distillation procedure (Hinton, 2015) lets us combine the accuracy of a large “teacher” model with the efficiency of a smaller “student” model.⁹

The two-step design has two practical advantages. First, although GPT-4 Turbo closely aligns with human judgments in validation tests, classifying roughly 100 million messages with GPT-4 Turbo would cost more than \$500,000; a fine-tuned BERT model is orders of magnitude cheaper. Second, our approach outperforms alternatives such as a purely dictionary-based method or a one-step design that trains BERT directly on self-declared investment approaches. Appendix Section A.1 reports these comparisons.

We apply the procedure separately to technical analysis (TA) and fundamental analysis (FA). $Usage^{TA}$ is an indicator equal to one if a message has a TA adoption probability above 95%, and $Usage^{FA}$ is the analogous indicator for FA.¹⁰ Section 6.3 extends the analysis to a third class, “other strategies” (OS), defined as messages that reference an investment strategy distinct from TA and FA.

TA Classification. We illustrate our classification procedure in detail for technical analysis. We first randomly sample 20,000 messages, drawing 10,000 from users with a self-declared technical approach and 10,000 from other users. For each message, we then prompt GPT-4

⁸Like Twitter (now X), StockTwits users post short messages: the original 140-character limit was expanded to 1,000 characters on May 8, 2019.

⁹See Gu et al. (2023) for a review of knowledge distillation in LLMs. We use the *bert-base-uncased* model (110 million parameters), substantially smaller than GPT-4 Turbo (1.7 trillion parameters). BERT is a well-established tool for classification tasks (Devlin et al., 2018), and González-Carvajal and Garrido-Merchán (2020) show that it consistently outperforms traditional NLP tools that do not rely on deep learning.

¹⁰TA and FA indicators are almost mutually exclusive because only 0.4% of messages exceed the 95% threshold for both, even though the two classifiers operate independently.

Turbo with:

You have a deep understanding of the language of social media and financial markets. Please analyze the message from an investor social media platform. Please parse the message along two dimensions. 1) Presence of technical analysis (0=no, 1=possibly, 2=likely). 2) If technical analysis is used, what is the technical indicator? (output the indicator or "" if you cannot locate it. If multiple signals exist, please separate by a comma) Output in JSON format: {"technical_analysis":, "technical_indicator": }.

Appendix Table A.1 reports examples of GPT responses from this prompt.¹¹

We then fine-tune a BERT classifier (“TA-BERT”) on these GPT outputs, where we treat GPT responses 1 and 2 as positive examples and response 0 as negative. TA-BERT achieves an F1 score of 0.83 in cross-validation, and its far smaller size allows local deployment on all 96 million messages.

Figures A.1 and A.2 plot the distribution of predicted TA adoption probabilities across investor groups by self-declared investment approach and investment horizon, respectively. Two features stand out. First, the TA-BERT predictions are bimodal: most messages fall either in the very low-probability region (< 5%) or in the very high-probability region (> 95%), so the classifications are typically unambiguous. Second, the predictions broadly line up with self-declarations – self-declared technical or momentum traders, and self-declared day or swing traders, post a substantially higher share of TA-related messages than other groups. The alignment is far from perfect, however: many self-declared fundamental, value, growth, or global-macro investors also post TA-related messages, illustrating the strategy malleability we analyze in Section 3.

Figure 2 presents word clouds of the most frequent unigrams and bigrams among TA-related messages. Panel A shows that technical messages frequently mention charts, in line with Jiang et al. (2023), and include canonical TA terms such as “resistance,” “support,” and “gap.” Panel

¹¹As a benchmark, we also implement a bag-of-words (BoW) approach using a TA-related keyword list from Cookson and Niessner (2020) and TF-IDF scores computed on the same 20,000 messages. The correlation is only 28% between BoW scores and GPT classifications. For 500 messages on which the two approaches disagree, a PhD research assistant manually classified each message based on her knowledge of technical analysis; the assistant’s labels agree with the GPT classifications 90% of the time.

B highlights bigrams that capture trading horizons, including “short-term” and “next week,” alongside other familiar TA expressions.

FA Classification. We also apply the same two-step procedure to fundamental analysis. The GPT-4 Turbo prompt becomes:¹²

You have a deep understanding of the language of social media and financial markets. Please analyze the message from an investor social media platform. Please parse the message along two dimensions. 1) Presence of fundamental analysis (0=no, 1=possibly, 2=likely). 2) If fundamental analysis is used, select one of the following 15 topics that is most relevant: "acquisitions-mergers", "analyst-ratings", "assets", "bankruptcy", "credit", "credit-ratings", "dividends", "earnings", "equity-actions", "investor-relations", "labor-issues", "marketing", "price-targets", "products-services", "revenues". Output in JSON format: {"fundamental_analysis":, "fundamental_topic":}.

We then fine-tune an FA-BERT classifier on the GPT outputs. Figures A.3 and A.4 plot the analogous adoption probability distributions for FA. As with TA-BERT, the distributions are bimodal and line up with self-declarations: users who self-report fundamental, value, or growth approaches post a higher share of FA-related messages. Figure 3 presents the corresponding word clouds, which highlight FA terms such as “earnings,” “price target,” and “long term.”

We also employ the same procedure for identifying OS messages.

2.2.2. Measuring Message Sentiment

We measure message sentiment using the user-declared bullish or bearish flag whenever available. For messages without a self-reported label, we impute sentiment from the message text using a supervised classifier. We randomly select 100,000 messages with a user-declared sentiment label, fine-tune a BERT model to predict the bullish indicator, and apply the model to the unlabeled messages.¹³

¹²The 15 fundamental-topic categories follow Ravenpack’s topic classification of financial news articles.

¹³The classifier achieves an F1 score of 0.9 from an out-of-sample set based on user-labeled messages excluded from the training sample. We choose BERT rather than GPT for measuring sentiment to mitigate look-ahead

We also aggregate message-level sentiment to the firm-day-strategy level. For each strategy $s \in \{TA, FA\}$ and stock i on day t , we focus on messages classified as a specific strategy type and, following [Cookson et al. \(2024\)](#), compute $Sentiment_{i,t}^s$ as the normalized difference between bullish and bearish messages:

$$Sentiment_{i,t}^s = \frac{N_{i,t}^{s,Bullish} - N_{i,t}^{s,Bearish}}{N_{i,t}^{s,Bullish} + N_{i,t}^{s,Bearish}}, \quad s \in \{TA, FA\}.$$

We set $Sentiment_{i,t}^s = 0$ on stock-days without any strategy s messages.

2.3. Other Variables

Retail trading. We construct two measures of net retail order imbalance, $OIB_{i,t}^{BJZZ}$ and $OIB_{i,t}^{BHJOS}$, based on the algorithms of [Boehmer et al. \(2021\)](#) (BJZZ) and [Barber et al. \(2023a\)](#) (BHJOS). To capture aggressive retail buying episodes, we follow [Barber et al. \(2022\)](#) and use RobinTrack data to construct a stock-day indicator for Robinhood herding, $RH\ Herd_{i,t}$, which equals one if stock i is among the top ten stocks ranked by the daily percentage increase in the number of Robinhood users holding the stock, conditional on at least 100 users holding the stock at the end of the prior day.

Firm characteristics. We control for the standard set of firm characteristics: *Market Capitalization*, *Book-to-Market*, *Asset Growth*, *Gross Profit-to-Assets*, *Analyst Coverage* (number of IBES equity analysts), and *Institutional Ownership* (the fraction of shares outstanding held by 13F institutions). *MAX* is the maximum one-day return in the preceding month, capturing lottery-like return patterns ([Bali et al., 2011](#)). *Abnormal Turnover* is the log difference between current daily turnover and the average over the prior four trading days, and *Abnormal News Volume* is the analogous log difference for the daily count of RavenPack news articles.

Attention. Following [Cookson et al. \(2024\)](#), we measure retail investor attention, $Attention_{i,t}$, as the number of StockTwits messages on stock i on day t scaled by the total number of messages posted that day.

bias. GPT models are trained on broad corpora that may include forward-looking financial text (see, e.g., [Sarkar and Vafa, 2024](#)), whereas BERT’s pretraining corpus (books and Wikipedia) is unlikely to contain such material ([Google BERT team, 2018](#)). Appendix Table A.7 confirms that our main results are robust to using the sentiment measure estimated from the maximum entropy model (see, [Cookson and Niessner, 2020](#)).

Table A.2 provides full construction details for these variables.

2.4. Summary Statistics

Table 1 reports summary statistics for the two main samples. Panel A corresponds to the full corpus of 96 million StockTwits messages. In total, approximately 32% of these messages refer to strategies. About 9% of messages are classified as TA, 14% as FA, and 9% as OS. Among the approximately 32 million messages from users with self-reported profile information, 34% come from self-identified technical investors and 13% from fundamental investors; 59% report a swing or day-trading horizon and 23% a long-term horizon; 30% report professional trading experience and 16% report novice experience.

We aggregate the message-level sentiment into stock-day level observations following the procedure in subsection 2.2.2. For firm-days without a valid message, we assign the sentiment score to 0. Panel B corresponds to approximately 3 million stock-day observations in the firm-day sample. Average $Sentiment^{TA}$ is 0.23 and average $Sentiment^{FA}$ is 0.29, both positive, indicating that the typical strategy-related message is bullish. Net retail order imbalance averages near zero under both identifications (-0.01 for BJZZ and BHJOS), with cross-sectional standard deviations of 0.27 and 0.25. The median stock has a market capitalization of \$1.2 billion (comparable to Giannini et al. 2018), is covered by seven equity analysts, and has 13F institutional ownership of 69%.

3. How Investors Choose Strategies

In this section, we examine how the strategy revealed in a StockTwits message varies across users and over time, focusing on Technical Analysis (TA) and Fundamental Analysis (FA). We first ask whether self-reported profile attributes explain strategy variation, and then test whether strategy choice responds to the information environment, past strategy performance, and peer feedback.

3.1. User Attributes and Firm News

We first ask how much of the variation in strategy choices is explained by self-reported traits. We estimate the following message-level regression on the sample of users with valid self-reported profile information:

$$Usage_{j,i,t,n}^s = \beta_1 X_j^{user} + \delta_i + \eta_t + \epsilon_{j,i,t,n}, \quad s \in \{TA, FA\}, \quad (1)$$

where $Usage_{j,i,t,n}^s$ equals one if message n , posted by user j about stock i on day t , is classified as discussing strategy $s \in \{TA, FA\}$. The vector X_j^{user} contains three sets of self-reported attributes. The first set is investment approach: *Technical Investor* and *Fundamental Investor* equal one if the user's self-reported approach is *Technical* or *Fundamental*, respectively, and the omitted approaches are *Momentum*, *Value*, *Growth*, and *Global Macro*. The second is holding horizon: *Swing or Day Trader* and *Long-Term Investor*, with *Position Trader* as the omitted category. The third is investment experience: *Novice* and *Professional*, with *Intermediate* as the omitted category. δ_i and η_t denote stock and date fixed effects, respectively.

Although self-reported investment approaches tend to be static, the strategy a user actually discusses varies from message to message. Panel A of Table 2 reports the regression results.¹⁴ Columns (1) to (3) focus on TA usage ($Usage^{TA}$). Column (1) includes only the two approach indicators. Self-reported *Technical Investors* are 12.5 percentage points more likely to post a TA message than the omitted approach group, an 89% increase relative to the unconditional TA frequency of 14%, while the coefficient on *Fundamental Investor* is small and statistically insignificant. The adjusted R^2 , however, is only 3.1%, indicating that a user's static profile maps weakly into within-user strategy choice. Column (2) adds self-declared horizon and experience indicators. Self-reported *Swing or Day Traders* are significantly more likely to discuss technical signals than the omitted group (*Position Traders*), while *Long-Term Investors* are less likely to do so. Self-reported *Professionals* are more likely to post TA than the omitted group (*Intermediate*), and *Novices* are significantly less likely to do so. Adding stock and date fixed effects in column (3) leaves the prior results qualitatively unchanged.

¹⁴This analysis requires self-reported user characteristics. Appendix Table A.3 Panel A reports summary statistics for the subsample used in this analysis.

Columns (4) to (6) report the analogous results for FA usage. The qualitative pattern flips. In column (4), self-reported *Fundamental Investors* are 8.6 percentage points more likely to post FA-related content (a 51% increase relative to the unconditional FA frequency of 17%), while *Technical Investors* are 3.5 percentage points less likely (a 21% decrease). Users who self-report a long-term horizon or professional experience are also more likely to post FA messages. Across all six specifications, the adjusted R^2 remains low (3.1–6.0% for TA and 1.0–4.3% for FA), confirming that static profile attributes leave most of the within-user variation in strategy choice unexplained.

Having established that static profile attributes explain only a small share of strategy variation, we next examine time-varying drivers of within-user strategy choice, beginning with the information environment. Prior research finds that information flow shapes attention and belief formation (Bordalo et al., 2012; Cookson et al., 2024). We therefore ask whether value-relevant information, such as earnings releases, analyst recommendations, or business news articles, shifts users toward FA and away from TA.

To study the effect of news on strategy choices, we estimate the following panel regression for message n , posted by user j concerning stock i on day t :

$$Usage_{j,i,t,n}^s = \beta \text{Public News Releases}_{i,t} + \gamma X_j^{user} + \delta_i + \eta_t + \epsilon_{j,i,t,n}, \quad s \in \{TA, FA\}. \quad (2)$$

$Usage_{j,i,t,n}^s$ is the message-level strategy indicator defined in Eq. (1). $\text{Public News Releases}_{i,t}$ is a vector of three indicators for earnings news, analyst news (e.g., recommendations and price targets), and business news (e.g., credit ratings and labor issues) on stock i on day t , identified from RavenPack. The control vector X_j^{user} consists of the same self-reported user attributes used in Eq. (1). In an alternative specification, we instead control for user fixed effects (ψ_j). δ_i and η_t denote stock and date fixed effects, respectively.

Panel B of Table 2 reports the regression results.¹⁵ Columns (1) and (3) include stock and date fixed effects and control for the self-reported user attributes from Eq. (1). Columns (2) and (4) saturate the specification with user fixed effects, so that the news coefficients are iden-

¹⁵We omit the control variables for brevity; Appendix Table A.13 presents the full specification with all controls. Note that this appendix table corresponds to the specification in Section 3.3, which represents the most saturated version of the strategy-choice regression.

tified from within-user variation in strategy choice on news days versus non-news days. All three news indicators are negatively associated with TA usage and positively associated with FA usage, and the signs are robust to the within-user specification. For example, in column (1), earnings news lowers TA usage by 3.2 percentage points (23% of the TA mean), and in column (3), it raises FA usage by 5.6 percentage points (33% of the FA mean). The magnitudes attenuate to 2.0 and 4.5 percentage points (14% and 26% of their respective means) once user fixed effects are included in columns (2) and (4). Analyst and business news exhibit qualitatively similar patterns. When firm-specific fundamentals become salient through corporate disclosure or media coverage, the same user posts more FA-related content and less TA-related content.

3.2. *Performance Feedback*

We next examine how prior strategy performance shapes the strategy choices revealed in subsequent messages. The experience-based learning framework (Malmendier and Nagel, 2011; Barberis and Jin, 2023) predicts that investors adjust their decision rules in response to realized outcomes.¹⁶ If retail investors reweight strategies based on realized outcomes along with their previous sentiments, poor outcomes under one strategy should reduce subsequent reliance on that strategy and increase reliance on alternatives.

To test this mechanism, we augment Eq. (2) with indicators for the user’s recent strategy-specific performance:

$$Usage_{j,i,t,n}^s = \sum_{k \in \{TA, FA\}} \beta_1^k Strategy\ Performance_{j,t-1}^k + \beta_2 Public\ News\ Releases_{i,t} + \gamma X_{j,t} + \psi_j + \delta_i + \eta_t + \epsilon_{j,i,t,n}, \quad s \in \{TA, FA\}. \quad (3)$$

For each strategy k , $Strategy\ Performance^k$ comprises two indicators, $High\ Performance^k$ and $Low\ Performance^k$, that summarize the user’s relative ranking among StockTwits users on the abnormal returns realized from k -classified messages. We construct each user’s strategy performance in two steps: first, we compute the five-day DGTW-adjusted return following each

¹⁶Empirically, investors overweight personal experience (Kaustia and Knüpfer, 2008; Seru et al., 2010; Huang, 2019), extrapolate recent returns (see, e.g., Greenwood and Shleifer, 2014; Da et al., 2021), engage in positive feedback trading (DeLong et al., 1990), and become more confident following favorable performance (Daniel et al., 1998).

message (buying on positive sentiment and shorting on negative sentiment); second, we average these abnormal returns across the user’s k -classified messages over the trailing three-month window, excluding the most recent five trading days to avoid look-ahead bias. *High Performance* ^{k} (*Low Performance* ^{k}) equals one if the user’s past performance is at or above the 75th percentile (at or below the 25th percentile) of the pooled cross-sectional distribution of strategy- k performance, and zero otherwise. The control vector $X_{j,t}$ contains the user’s past-quarter TA and FA posting intensity and is added in columns (2) and (4) of Table 3; ψ_j , δ_i , and η_t denote user, stock, and date fixed effects, respectively. The sample is restricted to investors who posted at least one TA or FA message in the prior three months.¹⁷

Table 3 reports the results. Column (1) shows that retail investors reduce their TA-related posts by 0.6 percentage points (a 4.3% decrease relative to the unconditional TA frequency of 14%) following poor TA performance. Conversely, they switch toward TA by 0.4 percentage points following poor FA performance. In column (2), we further control for the user’s past-quarter TA and FA usage, and the results remain robust. Columns (3) and (4) report analogous patterns for FA usage. The response is asymmetric: users move away from underperforming strategies, while strong performance does not significantly increase subsequent usage of that strategy. These findings provide direct, real-time evidence that investors adjust decision rules based on realized outcomes, consistent with experience-based learning models (Malmendier and Nagel, 2011) and strategy-switching frameworks (Barberis and Jin, 2023).

3.3. Social Feedback

Beyond experience with investment performance, retail investors also adapt their decision rules in response to peer feedback. Social-learning models emphasize that peer validation amplifies and propagates investment narratives (Shiller, 2017; Hirshleifer, 2020). If social reinforcement influences strategy choice, then the likes a user receives should shape which strategies the user continues to discuss in subsequent messages.

¹⁷Appendix Table A.3 Panel B reports the statistics for the 4.7 million messages in the active-user sample, which is restricted to users who posted at least one TA or FA message in the prior quarter. Average past TA and FA implied performance over the trailing three months, measured as DGTW-adjusted five-day returns following each message, are close to zero (−0.37% and −0.08%). The mean number of likes received on past TA and FA messages is 254 and 203, with substantial right-skewness (medians of 44 and 56).

To test this mechanism, we augment Eq. (3) with strategy-specific feedback indicators:

$$Usage_{j,i,t,n}^s = \sum_{k \in \{TA, FA\}} \beta_1^k Social\ Feedback_{j,t-1}^k + \sum_{k \in \{TA, FA\}} \beta_2^k Strategy\ Performance_{j,t-1}^k + \beta_3 Public\ News\ Releases_{i,t} + \gamma X_{j,t} + \psi_j + \delta_i + \eta_t + \epsilon_{j,i,t,n}, \quad s \in \{TA, FA\}. \quad (4)$$

For each strategy k , $Social\ Feedback^k$ comprises two indicators, $High\ Likes^k$ and $Low\ Likes^k$. $High\ Likes^k$ ($Low\ Likes^k$) equals one if the total number of likes the user received on k -classified messages over the trailing three-month window (excluding the most recent five trading days) is at or above the 75th percentile (at or below the 25th percentile) of the pooled cross-sectional distribution of strategy- k likes, and zero otherwise. $Strategy\ Performance^k$ is defined as in Eq. (3); the control vector $X_{j,t}$, the fixed effects, and the sample restriction are the same as in the experience-based learning specification.

Table 4 reports the results. Column (1) shows that investors are 0.9 percentage points more likely to mention TA when their prior TA messages received many likes (a 6.4% increase relative to the unconditional TA frequency). Social feedback also induces switching: TA usage rises when FA messages receive few likes and falls when they receive many. In column (2), we further control for past-quarter TA and FA performance, and the result is robust. Columns (3) and (4) report analogous patterns for FA usage. These findings align with social-learning models (Shiller, 2017; Hirshleifer, 2020) in which peer validation propagates decision rules. The response to peer feedback is also stronger than the response to one's own past performance, suggesting that social reinforcement is especially influential in shaping retail strategy choice. Quantitatively, the effect of receiving few TA likes on TA usage (column (2) of Table 4, -0.82 pp) is approximately 95% larger in magnitude than the effect of poor TA performance (column (2) of Table 3, -0.42 pp), both estimated in the fully-controlled specification that includes user, stock, and date fixed effects.

4. Strategy Sentiment and Returns

Do StockTwits sentiments forecast future stock returns, and does the answer depend on the strategy behind the message? If social media discussions capture value-relevant insights not yet in prices, sentiment should be a positive predictor of subsequent returns. If instead these opinions are uninformative, no predictable pattern should emerge. If they reflect biases that temporarily distort prices, sentiment should be a negative predictor as dislocations unwind. We test these predictions by linking strategy-specific sentiment to subsequent returns and examining the role of social feedback.

4.1. Next-Day Returns

We begin by examining next-day return predictability using sentiment measures constructed separately from messages classified as TA or FA. We estimate the following stock-day predictive regression:

$$Return_{i,t+1} = \beta_1 Sentiment_{i,t}^s + \beta_2 Attention_{i,t} + \gamma X_{i,t} + \delta_t + \epsilon_{i,t+1}, \quad s \in \{TA, FA\}, \quad (5)$$

where $Return_{i,t+1}$ is stock i 's return on the next trading day, $Sentiment_{i,t}^s$ is the stock-day sentiment for strategy $s \in \{TA, FA\}$ defined in Section 2.2.2, and $Attention_{i,t}$ is StockTwits user attention to stock i on day t (Cookson et al., 2024). The control vector $X_{i,t}$ includes log market capitalization, log book-to-market, asset growth, gross profit-to-assets, log of one plus analyst coverage, log institutional ownership, the maximum daily return in the previous month (MAX), abnormal turnover, abnormal news volume, and returns at five daily lags; δ_t denotes date fixed effects. Standard errors are clustered by trading day.

Table 5 reports the regression results, which reveal substantial heterogeneity in return predictability across strategy types.¹⁸ Columns (1) and (2) consider strategy-specific sentiment separately. TA sentiment is a significant negative predictor of next-day returns: stocks with more bullish TA messages underperform the following day. The economic magnitude is meaningful. Stocks with the most bearish TA sentiment ($Sentiment^{TA} = -1$) outperform those with

¹⁸See Appendix Table A.14 for the same table exhibiting full controls.

the most bullish ($Sentiment^{TA} = +1$) by about 3.2 bps the next day, the product of the regression coefficient (1.61 bps) and the two-unit range from -1 to $+1$.

In sharp contrast, FA sentiment significantly and *positively* predicts returns: stocks with the most bullish FA sentiment outperform those with the most bearish by about 2.7 bps, the product of the FA coefficient (1.37 bps) and the same two-unit range.

Column (3) reports a regression that includes both sentiment measures simultaneously. The coefficients on TA and FA sentiment retain their signs and statistical significance, with magnitudes similar to, and slightly larger than, the univariate estimates in columns (1) and (2).

4.2. Long-Short Portfolio Returns

To quantify the economic magnitude of the return predictability documented in the previous section, we construct long-short (L/S) portfolios using the entire cross-section of stocks with valid sentiment measures. Following Nagel (2005), we first residualize sentiment each day by estimating a cross-sectional regression of sentiment on a set of stock characteristics, including attention, the log of market capitalization, abnormal turnover, and five daily return lags. We then construct the L/S portfolio following Jensen et al. (2023): positions are taken in proportion to each stock’s residual sentiment. Specifically, the L/S portfolio is weighted by each stock’s residual sentiment deviation from the cross-sectional mean, and the portfolio return is calculated as:

$$r_t^{L/S} = \frac{\sum_{i=1}^N (S_{i,t-1} - \bar{S}_{t-1}) \times r_{i,t}}{\frac{1}{2} \sum_{i=1}^N |S_{i,t-1} - \bar{S}_{t-1}|}, \quad \text{where } \bar{S}_{t-1} = \frac{1}{N} \sum_{i=1}^N S_{i,t-1}.$$

For interpretability, TA sentiment is multiplied by -1 to reflect its negative relation to returns.

Panel A of Table 6 reports annualized average daily returns and Sharpe ratios. Consistent with the predictive regressions in Table 5, the strategy that bets against retail TA sentiment (long stocks with low TA sentiment, short stocks with high TA sentiment) generates an annualized return of 9.50% (t -statistic of 2.91) and a Sharpe ratio of 0.86. The FA-based L/S strategy delivers an annualized return of 6.48% (t -statistic of 2.04) and a Sharpe ratio of 0.58. The results are robust and somewhat stronger when we use DGTW-adjusted returns (Daniel et al.,

1997): Panel B shows that the TA (FA) L/S strategy yields an annualized DGTW-adjusted return of 10.10% (7.75%) and an information ratio of 1.00 (0.75).

These results confirm that the heterogeneity documented in the predictive regressions is economically meaningful. The contrarian signal in TA sentiment and the directional signal in FA sentiment both generate sizable abnormal returns. Although such strategies may not be implementable in practice given high turnover and transaction costs, the magnitudes help quantify how strongly retail sentiment relates to subsequent stock returns.

4.3. *Longer Horizon Returns*

To test whether our findings extend beyond the very short run, we estimate the predictive regression using cumulative stock returns over three windows: $t + 2$ to $t + 5$, $t + 6$ to $t + 10$, and $t + 11$ to $t + 15$. The results are reported in Table 7.

TA sentiment continues to negatively predict cumulative returns at both the short window ($t+2$ to $t+5$) and the intermediate window ($t+6$ to $t+10$), with point estimates of -5.36 (t -stat = -3.66) and -6.07 (t -stat = -3.76). These magnitudes are several times larger than the next-day effect. By $t + 11$ to $t + 15$, the predictability fades, with no evidence of a subsequent rebound. FA sentiment significantly predicts positive returns only over the short window ($+2.30$; t -stat = 1.88) and loses statistical significance at longer horizons.

These patterns further distinguish the informational content of the two strategies. FA sentiment correctly predicts near-term returns, consistent with the incorporation of value-relevant information. The negative predictability of TA sentiment, by contrast, persists for roughly two weeks before dissipating, consistent with TA-related signals reflecting behavioral biases that take time to correct. Investors who trade on TA sentiment therefore bear the cost of these reversals over the following one-to-two-week window, while those who follow fundamental discussions are positioned closer to the eventual price adjustment.

4.4. *Social Feedback and Sentiment Informativeness*

Our earlier analyses show that investors' strategy choices are significantly affected by social feedback. Whether such social interactions also influence the informativeness of the underlying

sentiment, however, is less clear. The literature paints a nuanced picture of social learning: Han et al. (2022) and Hirshleifer et al. (2025) show that social transmission leads to the adoption of suboptimal active strategies and to a preference for lottery-like stocks, while Pool et al. (2015) document settings in which social learning transmits useful information and helps fund managers improve performance.

We ask whether a strategy’s popularity among retail investors relates to its informativeness. To this end, we augment the predictive regression in Eq. (5) with strategy-specific likes and their interactions with sentiment:

$$\begin{aligned} Return_{i,h} = & \sum_{s \in \{TA, FA\}} [\beta_1^s Sentiment_{i,t}^s \times \text{Log}(1 + Likes_{i,t}^s) + \beta_2^s Sentiment_{i,t}^s + \beta_3^s \text{Log}(1 + Likes_{i,t}^s)] \\ & + \beta_4 Attention_{i,t} + \gamma X_{i,t} + \delta_t + \epsilon_{i,h}. \end{aligned} \quad (6)$$

$Likes_{i,t}^s$ is the total number of likes received by strategy- s messages on stock i on day t , and $Return_{i,h}$ is the cumulative return over forecast horizon h ($t + 1$, $t + 2$ to $t + 5$, $t + 6$ to $t + 10$, or $t + 11$ to $t + 15$); the remaining controls are as in Eq. (5).

Table 8 presents these results, revealing a stark divergence in how popularity affects the informativeness of sentiment across strategy types. When the TA strategy is highly popular, the negative correlation between TA sentiment and future returns strengthens. In sharp contrast, high FA popularity strengthens the positive correlation between FA sentiment and future returns. Social feedback, therefore, functions differently across strategies. For TA, it amplifies noise and likely exacerbates short-run mispricing; for FA, it manifests as the “wisdom of crowds,” with the most-liked messages carrying the strongest information about future returns.

The four sets of results in this section deliver a consistent message: strategy-specific sentiment is informative about future returns, but in opposite directions for the TA and FA approaches. Social validation pushes TA sentiment further toward noise but pushes FA sentiment closer to the truth. The signal extracted from retail commentary, therefore, depends crucially on the type of analysis behind the message.

5. Strategy Sentiment and Retail Trading

Having established that strategy-specific sentiment forecasts returns, we next ask whether these opinions are also linked to retail trading. If TA sentiment reflects unanchored signals spread through social interaction, it should show up most clearly in settings with crowded retail demand and subsequent reversals. If FA sentiment reflects value-relevant information, it should be less tied to such trading frenzies. We examine this link through intense Robinhood buying, short squeezes, and aggregate retail order imbalance.

5.1. *Intense Robinhood Buying*

[Barber et al. \(2022\)](#) find that intense buying by retail investors can produce significant temporary price surges and reversals. This setting is useful for our analysis because Robinhood investors are a visible group of retail traders whose behavior is closely tied to attention and social-media discussion. Robinhood users are disproportionately first-time or relatively inexperienced investors,¹⁹ and their trades are highly attention-driven. If socially amplified TA sentiment reflects unanchored signals, it should be especially likely to predict this type of concentrated buying. FA sentiment, by contrast, should be less connected to such trading episodes if it reflects value-relevant information.

Following [Barber et al. \(2022\)](#) and [Welch \(2022\)](#), we use RobinTrack data from May 2018 to August 2020 to identify intense-buying (buy-herding) episodes. We define an intense-buying episode for stock i on day t as one in which i ranks among the top ten stocks by the daily percentage increase in Robinhood users holding the stock, conditional on at least 100 users holding the stock at the end of day $t - 1$.

We first ask whether strategy-specific sentiment predicts these intense-buying episodes. In Section 4, social feedback strengthens the negative return predictability of TA sentiment but strengthens the positive return predictability of FA sentiment. The same distinction suggests a trading prediction: popular TA sentiment should be associated with crowded buying, whereas popular FA sentiment need not be. To test this prediction, we estimate the following stock-day

¹⁹For example, see <https://newsroom.aboutrobinhood.com/robinhood-raises-280-million-in-series-f-funding-led-by-sequoia/>.

panel regression:

$$RH_Herd_{i,t+h} = \sum_{s \in \{TA, FA\}} \left[\beta_1^s Sentiment_{i,t}^s \times \log(1 + Likes_{i,t}^s) + \beta_2^s Sentiment_{i,t}^s + \beta_3^s \log(1 + Likes_{i,t}^s) \right] + \beta_4 RH_Herd_{i,t} + \beta_5 Attention_{i,t} + \gamma X_{i,t} + \delta_t + \epsilon_{i,t+h}, \quad (7)$$

where the dependent variable is an indicator for whether stock i experiences intense buying on day $t + h$.

Table 9 reports the results across, with intense retail buying episodes over days $t+1$ to $t+3$ as the dependent variables in columns (1) through (3), respectively. The interaction of sentiment and likes is positive and significant for TA over the next two days, but negative and significant for FA. This pattern indicates that social feedback strengthens the positive link between TA sentiment and future intense buying, consistent with popular TA signals fostering crowded trading. By contrast, although FA sentiment itself positively predicts buy herding episodes, this link is substantially attenuated when FA messages receive more likes, suggesting that followers of FA strategies are less likely to move into crowded trades. This asymmetry is consistent with Stein (2009)'s distinction between unanchored strategies (such as momentum or technical signals, which spread crowded trading) and anchored ones (tied to fundamentals, which do not). Technical signals appear more likely to spread into crowded trades. Fundamental signals appear less likely to do so.

We next ask whether these intense-buying episodes lead to larger reversals when they occur alongside bullish TA sentiment. Prior work documents that intense Robinhood buying herding predicts low future returns. If TA sentiment helps identify the behavioral part of retail demand, then herding episodes should be especially costly when TA sentiment is high. We estimate the following panel regression:

$$Return_{i,h} = \sum_{s \in \{TA, FA\}} \left[\beta_1^s RH_Herd_{i,t} \times Sentiment_{i,t}^s + \beta_2^s Sentiment_{i,t}^s \right] + \beta_3 RH_Herd_{i,t} + \beta_4 Attention_{i,t} + \gamma X_{i,t} + \delta_t + \epsilon_{i,h}. \quad (8)$$

$RH_Herd_{i,t}$ is the herding indicator defined above, and the remaining controls follow Table 5.

Table A.4 reports the results. Column (1) replicates the baseline specification of Barber

et al. (2022): $RH_Herd_{i,t}$ predicts negative next-day returns, with a point estimate of -75 bps, confirming that herding episodes are followed by negative returns. Column (2) finds that the reversal is substantially stronger with higher TA sentiment. The interaction between TA sentiment and the herding indicator is negative and statistically significant, with a coefficient of -159 bps and a t -statistic of -3.20 . On herding days that coincide with the most bullish TA sentiment, the implied next-day reversal is 1.7 times $(= (28.38 - 159.41)/(-75.48))$ the baseline herding reversal magnitude. The corresponding interaction with FA sentiment is not statistically significant. Column (3) extends the horizon to $t+2$ through $t+5$, where the TA interaction remains negative but is not statistically significant. The evidence suggests that crowded buying linked to technical discourse is especially prone to reversal, whereas fundamental discussions do not display the same reversal amplification.

5.2. *Short Squeeze*

Short squeezes provide a sharper test of crowded retail demand. A squeeze combines a large price increase with short covering, so it captures both buying pressure and the forced unwinding of short positions. These episodes are important because they represent large price dislocations. They can impose losses not only on retail investors who buy near the peak, but also on sophisticated investors who are forced to cover short positions. Schultz (2024) documents that short squeezes are often followed by significant return reversals. Allen et al. (2025) find that social-media coordination can help trigger squeezes in heavily shorted stocks. The GameStop and AMC episodes of early 2021 made this channel salient. We ask whether the same TA-versus-FA contrast also appears in these more extreme episodes of crowded trading.

We follow Schultz (2024) and define a short-squeeze event for stock i on day t as a stock-day in which the one-day stock return exceeds 30% and the daily percentage change in short loan value decreases by more than 20%. We identify 1,099 short-squeeze events concerning 883 unique firms in our sample.

To test whether socially amplified sentiment predicts squeezes, we estimate a regression

analogous to Eq. (7), replacing the herding indicator with $Squeeze_{i,t+h}$:

$$Squeeze_{i,t+h} = \sum_{s \in \{TA, FA\}} \left[\beta_1^s Sentiment_{i,t}^s \times \log(1 + Likes_{i,t}^s) + \beta_2^s Sentiment_{i,t}^s + \beta_3^s \log(1 + Likes_{i,t}^s) \right] + \beta_4 Attention_{i,t} + \gamma X_{i,t} + \delta_t + \epsilon_{i,t+h}. \quad (9)$$

Table 10 reports the results for $h = 1, 2, 3$. The pattern mirrors the Robinhood evidence. Popular TA sentiment predicts future short squeezes, while popular FA sentiment does not. The interaction between TA sentiment and TA likes is positive and statistically significant at all three horizons. The corresponding FA interaction is small and statistically insignificant. TA likes also predict squeezes on their own, consistent with the idea that active technical discussion helps concentrate attention and buying pressure in heavily shorted stocks.

We then ask whether TA sentiment also predicts larger post-squeeze reversals. If bullish TA sentiment captures crowded demand, squeezes with high TA sentiment should reverse more sharply. Table A.5 reports the results. Column (1) confirms the squeeze-reversal pattern of [Schultz \(2024\)](#): squeeze days are followed by -602 bps next-day returns, an order of magnitude larger than the corresponding effect for Robinhood herding. Column (2) adds the sentiment interactions. The TA sentiment interaction is negative, implying larger reversals when squeeze days coincide with bullish TA sentiment: squeezes accompanied by the most bullish TA sentiment reverse roughly 40% more than the typical squeeze ($= -211/-505$). Column (3) shows that the TA interaction becomes stronger over the $t + 2$ to $t + 5$ horizon.

The squeeze evidence reinforces the Robinhood results. The same TA-versus-FA contrast appears in a different and more extreme setting: price runups with short covering. This pattern also sharpens the social-media squeeze mechanism in [Allen et al. \(2025\)](#). Not all retail chatter predicts squeezes. The predictive content is concentrated in technical signals that receive social engagement. Fundamental discussion, even when popular, does not produce the same squeeze-and-reversal pattern.

5.3. Aggregate Retail Order Flow

The Robinhood and squeeze tests focus on specific episodes of crowded retail demand. We now broaden the analysis to aggregate retail trading. Following [Boehmer et al. \(2021\)](#) and [Barber et al. \(2023a\)](#), we identify retail market orders and construct two measures of net retail order imbalance, $OIB_{i,t}^{BJZZ}$ and $OIB_{i,t}^{BHJOS}$. We estimate the following stock-day panel regression:

$$OIB_{i,t} = \sum_{s \in \{TA, FA\}} \beta_1^s \text{Sentiment}_{i,t}^s + \beta_2 \text{Attention}_{i,t} + \gamma X_{i,t} + \delta_t + \epsilon_{i,t}. \quad (10)$$

To align the sentiment measures with intraday retail order flow, $\text{Sentiment}_{i,t}^s$ in this regression is constructed using only messages posted during regular trading hours (9:30–16:00 ET). The control vector $X_{i,t}$ is the same as in Table 5, and δ_t denotes date fixed effects.

Table 11 reports the results. Both TA and FA sentiment are significantly positively associated with contemporaneous retail order imbalance. A one-standard-deviation increase in TA sentiment corresponds to a 0.46 percentage-point increase in net retail buying under the BJZZ measure, and a one-standard-deviation increase in FA sentiment corresponds to a 0.38 percentage-point increase. The pattern is similar using the BHJOS measure. Thus, strategy-specific opinions on StockTwits are directly linked to retail trading.²⁰

The key question is whether these different parts of retail trading have the same return implications. To answer this question, we decompose aggregate retail order imbalance into the portions linked to TA and FA sentiment. For each algorithm $j \in \{BJZZ, BHJOS\}$, we use the fitted values from Eq. (10) to construct a TA-driven component, an FA-driven component, and a residual component of retail order imbalance. We then replace strategy-specific sentiment in Eq. (5) with these three order-flow components and predict future returns across the four horizons: [0, 1], [2, 5], [6, 10], and [11, 15]:

$$\text{Return}_{i,h} = \beta_1 OIB_{i,t}^{TA} + \beta_2 OIB_{i,t}^{FA} + \beta_3 OIB_{i,t}^{Resid} + \beta_4 \text{Attention}_{i,t} + \gamma X_{i,t} + \delta_t + \epsilon_{i,h}, \quad (11)$$

The remaining controls follow Eq. (5).

Table 12 reports a clear contrast. The TA-linked component of retail order imbalance neg-

²⁰The R^2 values of these regressions are roughly 1%, indicating that retail order flow has many drivers other than the strategy-specific sentiment we measure here.

actively predicts future returns. The FA-linked component positively predicts future returns at short horizons. The residual component is positively related to future returns, consistent with [Boehmer et al. \(2021\)](#). The coefficient on FA-driven OIB is 3.90%, roughly 23 times that on the residual component (0.17%), consistent with FA-linked order flow being substantially more informed. The coefficient on TA-driven OIB is -3.12%, roughly 18 times the residual in magnitude but with the opposite sign. This pattern is consistent with TA-linked order flow capturing speculative demand driven by social media amplification of unanchored technical signals, pushing prices above fundamental value and generating reversals as the mispricing corrects.

Our findings help explain why aggregate retail order flow can look informed in some settings and behavioral in others. Retail order flow is not monolithic. The part linked to FA discussions appears informed, while the part linked to TA discussions appears behavioral. Together with the Robinhood herding and short-squeeze results, the evidence suggests that conditioning on social-media strategy discourse helps distinguish informed from speculative retail demand across both aggregate order flow and crowded trading episodes.

6. Robustness and Extensions

We supplement the main analyses with a set of robustness checks and additional results. Section [6.1](#) establishes that our findings are robust to alternative subsamples, sentiment-construction choices, and return measures. Section [6.2](#) extends the return-predictability evidence by examining how sentiment informativeness varies with the sophistication of the posting users. Section [6.3](#) introduces a third class of messages, “other strategies” (OS), and shows that the TA-versus-FA contrast survives the inclusion of OS as a control. OS itself behaves as a speculative residual, with contemporaneous order-flow correlations and subsequent return reversals that mirror TA rather than FA.

6.1. Robustness

Subsamples. We document the stability of our results across two subperiods (2010–2017 and 2018–2023) and three return horizons (next-day, $t+2$ to $t+5$, and $t+6$ to $t+10$). The 2018–2023 subperiod coincides with the rapid growth of retail trading and the rise of investment-focused social media, both of which arguably make the sentiment we extract from StockTwits a tighter measure of representative retail beliefs. Table A.6 re-estimates the predictive regression in Eq. (5) for each combination of subperiod and horizon, giving six columns. The data reveal a horizon-dependent shift in predictability. At the immediate next-day horizon ($t+1$), both the negative predictability of technical (TA) sentiment and the positive predictability of fundamental (FA) sentiment are concentrated entirely in the 2018–2023 subperiod. The coefficients in the 2010–2017 subperiod are near zero and statistically insignificant. As the horizon extends beyond the next day, FA predictability becomes insignificant in both subperiods, while TA predictability persists. At the $t+2$ to $t+5$ horizon, negative TA predictability is significant in both periods, though its magnitude roughly doubles during 2018–2023 (−6.84 versus −3.38). At the longest horizon ($t+6$ to $t+10$), negative TA predictability remains strongly significant across both subperiods. Overall, this pattern aligns with our interpretation: as retail participation deepens and accelerates, StockTwits sentiment more accurately reflects the beliefs of the active retail trader, strengthening its immediate, short-term transmission into stock returns.

Alternative Sentiment Measure. We probe the robustness of our findings to the choice of sentiment measure by replacing the BERT-based sentiment classification with the rule-based algorithm of Cookson and Niessner (2020), which infers message sentiment from a curated lexicon of bullish and bearish terms. Table A.7 reports the results. The pattern from Table 5 survives: TA sentiment continues to negatively predict next-day returns, FA sentiment continues to positively predict them, and the magnitudes are comparable to the baseline estimates. The predictability we document, therefore, does not hinge on BERT.

Fama-MacBeth Estimation. We examine whether our findings are sensitive to the choice of estimator. The pooled-panel specifications with date fixed effects in our main tables exploit

cross-sectional variation across stocks within day, but the resulting standard errors may underweight time-series variation in the estimated relationships. Table A.8 re-estimates the return-predictability regression using the Fama-MacBeth approach: we estimate the cross-sectional specification in Eq. (5) each day and report the time-series average of the estimated coefficients, with Newey-West standard errors (one lag) on the resulting time series. The TA-versus-FA pattern is preserved. FA sentiment is significantly positive at the next-day horizon (1.31 bps, $t = 2.20$), and TA sentiment is significantly negative at the two intermediate horizons (-4.71 bps at $t + 2$ to $t + 5$, $t = -3.71$; -5.68 bps at $t + 6$ to $t + 10$, $t = -3.99$).

Alternative Return Measure. We examine the sensitivity of our findings to the choice of return measure. Table A.9 replaces raw next-day returns with DGTW-adjusted abnormal returns (Daniel et al., 1997), which control for systematic differences in returns across size, book-to-market, and momentum-sorted portfolios. The DGTW-adjusted estimates closely mirror the baseline, indicating that the predictability we document is not driven by systematic exposure to size, value, or momentum factors.

6.2. *Investor Sophistication*

We examine whether the informativeness of StockTwits sentiment varies with investor sophistication. Although TA sentiment is on average negatively related to subsequent returns, prior research finds that certain technical signals can be profitable (e.g., Jegadeesh and Titman, 1993; Han et al., 2013, 2016; Jiang et al., 2023; Murray et al., 2024). This raises the possibility that TA sentiment is less behaviorally biased when sophisticated participants account for a larger share of the conversation.

We measure sophistication, $FracProMsg_{i,t}$, as the fraction of messages on stock i on day t contributed by self-reported professional investors. We then augment the predictive regression in Eq. (5) with $FracProMsg_{i,t}$ and its interaction with strategy-specific sentiment, estimated separately for cumulative returns at four horizons ($t + 1$, $t + 2$ to $t + 5$, $t + 6$ to $t + 10$, and $t + 11$ to $t + 15$). Table A.10 shows that the interaction between $FracProMsg_{i,t}$ and TA sentiment is positive and significant at every horizon, with point estimates rising from 3.03 bps at $t + 1$

to a peak of 13.48 bps over $t + 6$ to $t + 10$ before partially fading at $t + 11$ to $t + 15$. As professional voices grow more prominent, the otherwise negative predictability of TA sentiment is attenuated, consistent with the view that professional participation tempers the behavioral component of technical-style speculation. The predictive power of FA sentiment, by contrast, is essentially unaffected by user composition at any horizon, consistent with a “wisdom of crowds” interpretation in which fundamental signals are informative even when the participant pool skews toward less sophisticated retail investors (Chen et al., 2014).

6.3. Other Strategies

The main analysis isolates TA and FA, the two strategy classes that comprise the bulk of strategy-related messages. A natural question is whether the patterns we document carry over to messages that invoke other investment strategies. We label these messages OS (“other strategies”), which together account for 28% of all strategy-related posts in our sample.

OS Classification. We construct OS using the same two-step LLM-then-BERT procedure described in Section 2.2.1. The GPT-4 Turbo prompt asks whether the message references any investment strategy, with examples spanning technical analysis, fundamental analysis, event-driven strategies, and arbitrage:

You have a deep understanding of the language of social media and financial markets. Please analyze the message from an investor social media platform. Please parse the message along two dimensions. 1) Presence of investment strategy (e.g., technical analysis, fundamental analysis, event-driven strategy, arbitrage strategy). If true, please answer 1, otherwise 0. 2) If a strategy is identified, please specify the strategy. Output in JSON format: {“has_strategy”: , “strategy_type”: }.

We then fine-tune a BERT model on these GPT outputs and set $Usage_{j,i,t,n}^{OS}$ to one when the BERT probability exceeds 95% and the message is not also classified as TA or FA. OS sentiment, $Sentiment_{i,t}^{OS}$, follows the same firm-day aggregation we use for TA and FA.

OS Strategy Choice. Table A.11 re-runs the strategy-choice regressions of Tables 2, 3, and 4 with $Usage_{j,i,t,n}^{OS}$ on the left-hand side. The four columns add layers of regressors: column (1) uses self-reported user attributes (Eq. (1)); column (2) replaces them with news indicators and adds user fixed effects (Eq. (2)); column (3) further adds past TA, FA, and OS performance (Eq. (3)); and column (4) further adds the past-likes counterparts (Eq. (4)).

The user-attribute coefficients in column (1) point to the speculative end of the user spectrum. *Technical Investors* ($t = 3.59$), *Swing or Day Traders* ($t = 5.45$), and *Professionals* ($t = 6.31$) post more OS messages, while *Fundamental Investors* ($t = -1.99$), *Long-Term Investors* ($t = -6.63$), and *Novices* ($t = -8.68$) post fewer. The same investor groups that lean toward TA also lean toward OS.

Column (2) shows that earnings, analyst, and business news are all significantly *negative* predictors of OS usage. The pattern matches that for TA in Table 2: when fundamental information arrives, attention shifts to FA and away from both TA and OS.

Past performance has little explanatory power for OS usage in column (3). Past likes, by contrast, induce a clear pattern of strategy substitution in column (4). When a user’s recent TA or FA messages receive few likes (*Low TA Likes*: $t = 2.96$; *Low FA Likes*: $t = 2.55$), the user shifts toward OS. When FA messages received many likes (*High FA Likes*: $t = -2.18$), the user shifts away from OS. OS, therefore, absorbs the posting effort that TA and FA do not reward.

OS Sentiment, Returns, and Order Flow. Table A.12 brings OS into our two main tests by adding OS sentiment as a third regressor alongside TA and FA. We also include $Sentiment_{i,t}^{NS}$, constructed from messages that fall outside the TA, FA, and OS labels, so that the four regressors span the full set of strategy-related and non-strategy-related messages. Column (1) re-estimates the next-day predictive regression in Eq. (5); columns (2) and (3) re-estimate Eq. (10) for the BJZZ and BHJOS retail-order measures.

The headline result is that $Sentiment_{i,t}^{OS}$ negatively predicts next-day returns, with the negative TA effect from Table 5 and the positive FA effect both surviving the inclusion of OS and NS as additional regressors. NS sentiment, by contrast, has limited predictive content for next-day returns once the strategy-specific terms are included.

The OIB columns tell a parallel story. OS sentiment is positively associated with con-

temporaneous retail order flow under both the BJZZ and BHJOS measures, with magnitudes comparable to TA and FA. OS-classified messages, therefore, translate into retail trading just as TA and FA messages do, but the resulting flow is followed by reversals, much like the flow associated with TA.

7. Conclusion

This study uses high-frequency social media data and LLM-based classification to study how retail investors form opinions. By combining large-scale social media data with LLM-based classification, we measure two key components of expressed opinions: the investment strategy referenced in a message and the sentiment expressed. We find that strategy use is dynamic rather than a fixed trait. The same investor shifts across strategies in response to firm news, past strategy performance, and social feedback.

We further document sharp heterogeneity in the informational content of expressed opinions. Sentiment in fundamental-analysis (FA) messages positively predicts future returns, whereas sentiment in technical-analysis (TA) and other-strategy (OS) messages predicts returns in the opposite direction, particularly when discussions are dominated by less sophisticated users. These strategy-specific sentiments are directly linked to retail trading. TA sentiment is most strongly linked to intense buying episodes on Robinhood and to short squeezes, both followed by sharp reversals. Decomposing retail order flow indicates that the component tied to FA-linked discussions correctly predicts future returns, while the components associated with TA and OS discussions are negatively predictive.

These patterns suggest that retail order flow is not monolithic. FA-linked discussions are consistent with a wisdom-of-crowds channel, in which social media aggregates value-relevant information. TA discussions, by contrast, are more consistent with social transmission of unanchored signals, which can generate crowded trading and reversals. This distinction helps clarify when retail participation contributes to price discovery and when it instead coincides with speculative demand.

Taken together, our evidence demonstrates that social media discourse provides a useful

window into belief updating and decision-making at high frequency. By observing both what investors think and the strategies they use to form those opinions, our approach helps open the black box of retail opinion formation.

REFERENCES

- Abarbanell, Jeffrey S, and Brian J Bushee, 1997, Fundamental analysis, future earnings, and stock prices, *Journal of accounting research* 35, 1–24.
- Allen, Franklin, Marlene Haas, Eric Nowak, Matteo Pirovano, and Angel Tenguov, 2025, Squeezing shorts through social media platforms, *Management Science* .
- Bali, Turan G, Nusret Cakici, and Robert F Whitelaw, 2011, Maxing Out: Stocks As Lotteries and The Cross-Section of Expected Returns, *Journal of financial economics* 99, 427–446.
- Barber, Brad M, Xing Huang, Philippe Jorion, Terrance Odean, and Christopher Schwarz, 2023a, A (Sub)penny for Your Thoughts: Tracking Retail Investor Activity in TAQ, *The Journal of Finance* .
- Barber, Brad M, Xing Huang, Terrance Odean, and Christopher Schwarz, 2022, Attention-Induced Trading and Returns: Evidence from Robinhood Users, *The Journal of Finance* 77, 3141–3190.
- Barber, Brad M, Shengle Lin, and Terrance Odean, 2023b, Resolving a Paradox: Retail Trades Positively Predict Returns but Are Not Profitable, *Journal of Financial and Quantitative Analysis* 1–35.
- Barber, Brad M, and Terrance Odean, 2000, Trading Is Hazardous to Your Wealth: The Common Stock Investment Performance of Individual Investors, *The journal of Finance* 55, 773–806.
- Barber, Brad M, and Terrance Odean, 2008, All That Glitters: The Effect of Attention and News on the Buying Behavior of Individual and Institutional Investors, *The Review of Financial Studies* 21, 785–818.
- Barberis, Nicholas C., and Lawrence J. Jin, 2023, Model-free and model-based learning as joint drivers of investor behavior, NBER Working Paper 31081, National Bureau of Economic Research.
- Barrot, Jean-Noel, Ron Kaniel, and David Sraer, 2016, Are retail traders compensated for providing liquidity?, *Journal of Financial Economics* 120, 146–168.
- Bastianello, Francesca, Paul Decaire, and Marius Guenzel, 2024, Mental models and financial forecasts, *Jacobs Levy Equity Management Center for Quantitative Financial Research Paper* .
- Bastianello, Francesca, and Paul Fontanier, 2025, Expectations and learning from prices, *The Review of Economic Studies* 92, 1341–1374.
- Blume, Lawrence, David Easley, and Maureen O’hara, 1994, Market Statistics and Technical Analysis: The Role of Volume, *The Journal of Finance* 49, 153–181.
- Boehmer, Ekkehart, Charles M Jones, Xiaoyan Zhang, and Xinran Zhang, 2021, Tracking Retail Investor Activity, *The Journal of Finance* 76, 2249–2305.
- Bordalo, Pedro, Nicola Gennaioli, and Andrei Shleifer, 2012, Salience theory of choice under risk, *The Quarterly Journal of Economics* 127, 1243–1285.
- Brock, William, Josef Lakonishok, and Blake LeBaron, 1992, Simple Technical Trading Rules and The Stochastic Properties of Stock Returns, *The Journal of Finance* 47, 1731–1764.
- Brown, David P, and Robert H Jennings, 1989, On Technical Analysis, *The Review of Financial Studies* 2, 527–551.
- Bryzgalova, Svetlana, Anna Pavlova, and Taisiya Sikorskaya, 2023, Retail Trading in Options and the Rise of the Big Three Wholesalers, *The Journal of Finance* 78, 3465–3514.
- Campbell, John Y, and Tarun Ramadorai, 2026, Household finance in retrospect and prospect, Technical report, National Bureau of Economic Research.
- Chen, Hailiang, Prabuddha De, Yu Hu, and Byoung-Hyoun Hwang, 2014, Wisdom of Crowds: The Value of Stock Opinions Transmitted Through Social Media, *The Review of Financial Studies* 27, 1367–1403.

- Chinco, Alex, Samuel M Hartzmark, and Abigail B Sussman, 2022, A New Test of Risk Factor Relevance, *The Journal of Finance* 77, 2183–2238.
- Choi, James J, and Adriana Z Robertson, 2020, What Matters to Individual Investors? Evidence from The Horse’s Mouth, *The Journal of Finance* 75, 1965–2020.
- Cookson, J Anthony, Joseph E Engelberg, and William Mullins, 2023, Echo Chambers, *The Review of Financial Studies* 36, 450–500.
- Cookson, J Anthony, Runjing Lu, William Mullins, and Marina Niessner, 2024, The Social Signal, *Journal of Financial Economics* 158, 103870.
- Cookson, J Anthony, and Marina Niessner, 2020, Why Don’t We Agree? Evidence from A Social Network of Investors, *The Journal of Finance* 75, 173–228.
- Da, Zhi, Xing Huang, and Lawrence J Jin, 2021, Extrapolative beliefs in the cross-section: What can we learn from the crowds?, *Journal of Financial Economics* 140, 175–196.
- Daniel, Kent, Mark Grinblatt, Sheridan Titman, and Russ Wermers, 1997, Measuring mutual fund performance with characteristic-based benchmarks, *The Journal of finance* 52, 1035–1058.
- Daniel, Kent, David Hirshleifer, and Avanidhar Subrahmanyam, 1998, Investor psychology and security market under- and overreactions, *The Journal of Finance* 53, 1839–1885.
- DeLong, J. Bradford, Andrei Shleifer, Lawrence H. Summers, and Robert J. Waldmann, 1990, Positive feedback investment strategies and destabilizing rational speculation, *Journal of Finance* 45, 375–395.
- Devlin, Jacob, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova, 2018, BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding, *CoRR* abs/1810.04805.
- Eaton, Gregory W, T Clifton Green, Brian S Roseman, and Yanbin Wu, 2022, Retail Trader Sophistication and Stock Market Quality: Evidence from Brokerage Outages, *Journal of Financial Economics* 146, 502–528.
- European Central Bank, November 2025, Financial stability review, november 2025, <https://www.ecb.europa.eu/press/financial-stability-publications/fsr/html/ecb.fsr202511~263b5810d4.en.html>, Accessed: 9 June 2026.
- Foucault, Thierry, David Sraer, and David J Thesmar, 2011, Individual investors and volatility, *The Journal of Finance* 66, 1369–1406.
- Gao, Zhenyu, Wei Xiong, and Jian Yuan, 2025, Structured beliefs in fund managers’ own words: Fundamentals, policy, and sentiment, Working paper, Princeton University.
- George, Thomas J, and Chuan-Yang Hwang, 2004, The 52-Week High and Momentum Investing, *The Journal of Finance* 59, 2145–2176.
- Giannini, Robert, Paul Irvine, and Tao Shu, 2018, Nonlocal Disadvantage: An Examination of Social Media Sentiment, *The Review of Asset Pricing Studies* 8, 293–336.
- Giglio, Stefano, Matteo Maggiori, Johannes Stroebel, and Stephen Utkus, 2021, Five Facts about Beliefs and Portfolios, *American Economic Review* 111, 1481–1522.
- González-Carvajal, Santiago, and Eduardo C Garrido-Merchán, 2020, Comparing BERT against Traditional Machine Learning Text Classification, *arXiv preprint arXiv:2005.13012*.
- Google BERT team, 2018, BERT Base Uncased Model, <https://huggingface.co/google-bert/bert-base-uncased>, Hugging Face Model Card; Accessed on 26 August 2025.
- Greenwood, Robin, and Andrei Shleifer, 2014, Expectations of returns and expected returns, *The Review of Financial Studies* 27, 714–746.

- Gu, Yuxian, Li Dong, Furu Wei, and Minlie Huang, 2023, Knowledge Distillation of Large Language Models, *arXiv preprint arXiv:2306.08543* .
- Han, Bing, David Hirshleifer, and Johan Walden, 2022, Social transmission bias and investor behavior, *Journal of Financial and Quantitative Analysis* 57, 390–412.
- Han, Bing, Haoyang Liu, and Pengfei Sui, 2023, Social learning and sentiment contagion in the bitcoin market, *Available at SSRN* 4543326.
- Han, Yufeng, Ke Yang, and Guofu Zhou, 2013, A New Anomaly: The Cross-Sectional Profitability of Technical Analysis, *Journal of Financial and Quantitative Analysis* 48, 1433–1461.
- Han, Yufeng, Guofu Zhou, and Yingzi Zhu, 2016, A Trend Factor: Any Economic Gains from Using Information over Investment Horizons?, *Journal of Financial Economics* 122, 352–375.
- Hinton, G, 2015, Distilling the Knowledge in A Neural Network, *arXiv preprint arXiv:1503.02531* .
- Hirshleifer, David, 2020, Presidential address: Social transmission bias in economics and finance, *The Journal of Finance* 75, 1779–1831.
- Hirshleifer, David, Lin Peng, and Qiguang Wang, 2025, News diffusion in social networks and stock market reactions, *The Review of Financial Studies* 38, 883–937.
- Huang, Xing, 2019, Mark twain’s cat: Investment experience, categorical thinking, and stock selection, *Journal of Financial Economics* 131, 404–432.
- Jegadeesh, Narasimhan, 1991, Seasonality in Stock Price Mean Reversion: Evidence from The US and The UK, *The Journal of Finance* 46, 1427–1444.
- Jegadeesh, Narasimhan, and Sheridan Titman, 1993, Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency, *The Journal of Finance* 48, 65–91.
- Jensen, Theis Ingerslev, Bryan Kelly, and Lasse Heje Pedersen, 2023, Is there a replication crisis in finance?, *The Journal of Finance* 78, 2465–2518.
- Jha, Manish, Jialin Qian, Michael Weber, and Baozhong Yang, 2024, Harnessing generative AI for economic insights, *Available at SSRN* 4976759 .
- Jha, Manish, Jialin Qian, Michael Weber, and Baozhong Yang, 2025, ChatGPT and Corporate Policies, Technical report, National Bureau of Economic Research.
- Jiang, Jingwen, Bryan Kelly, and Dacheng Xiu, 2023, (re-) Imag (in) ing Price Trends, *The Journal of Finance* 78, 3193–3249.
- Jiang, Zhengyang, Cameron Peng, and Hongjun Yan, 2024, Personality Differences and Investment Decision-Making, *Journal of Financial Economics* 153, 103776.
- Kaniel, Ron, Gideon Saar, and Sheridan Titman, 2008, Individual Investor Trading and Stock Returns, *The Journal of Finance* 63, 273–310.
- Kaustia, Markku, and Samuli Knüpfer, 2008, Do investors overweight personal experience? evidence from ipo subscriptions, *The journal of finance* 63, 2679–2702.
- Ke, Shikun, 2024, Analysts’ belief formation in their own words, *Available at SSRN* 5025830 .
- Kelley, Eric K, and Paul C Tetlock, 2013, How Wise Are Crowds? Insights from Retail Orders and Stock Returns, *The Journal of Finance* 68, 1229–1265.
- Kogan, Shimon, Igor Makarov, Marina Niessner, and Antoinette Schoar, 2024, Are cryptos different? evidence from retail trading, *Journal of Financial Economics* 159, 103897.

- Kumar, Alok, and Charles MC Lee, 2006, Retail Investor Sentiment and Return Comovements, *The Journal of Finance* 61, 2451–2486.
- Liu, Hongqi, Cameron Peng, Wei A Xiong, and Wei Xiong, 2022, Taming The Bias Zoo, *Journal of Financial Economics* 143, 716–741.
- Lo, Andrew W, Harry Mamaysky, and Jiang Wang, 2000, Foundations of Technical Analysis: Computational Algorithms, Statistical Inference, and Empirical Implementation, *The Journal of Finance* 55, 1705–1765.
- Malmendier, Ulrike, and Stefan Nagel, 2011, Depression babies: Do macroeconomic experiences affect risk taking?, *The Quarterly Journal of Economics* 126, 373–416.
- Martin, Katie, and Robin Wigglesworth, 2021, Rise of the retail army: the amateur traders transforming markets, *Financial Times*, Accessed 2026-02-19.
- Meeuwis, Maarten, Jonathan A. Parker, Antoinette Schoar, and Duncan Simester, 2022, Belief disagreement and portfolio choice, *Journal of Finance* 77, 3191–3247.
- Murray, Scott, Yusen Xia, and Houping Xiao, 2024, Charting by Machines, *Journal of Financial Economics* 153, 103791.
- Nagel, Stefan, 2005, Short sales, institutional investors and the cross-section of stock returns, *Journal of financial economics* 78, 277–309.
- Pedersen, Lasse Heje, 2022, Game on: Social networks and markets, *Journal of Financial Economics* 146, 1097–1119.
- Pew Research Center, 2025, Social media and news fact sheet.
- Pool, Veronika K, Noah Stoffman, and Scott E Yonker, 2015, The people in your neighborhood: Social interactions and mutual fund portfolios, *The Journal of Finance* 70, 2679–2732.
- Porta, Rafael La, Josef Lakonishok, Andrei Shleifer, and Robert Vishny, 1997, Good news for value stocks: Further evidence on market efficiency, *the Journal of Finance* 52, 859–874.
- Sarkar, Suproteem K, and Keyon Vafa, 2024, Lookahead bias in pretrained language models, *Available at SSRN 4754678*.
- Schultz, Paul, 2024, Short squeezes and their consequences, *Journal of Financial and Quantitative Analysis* 59, 68–96.
- Seru, Amit, Tyler Shumway, and Noah Stoffman, 2010, Learning by trading, *The Review of Financial Studies* 23, 705–739.
- Shiller, Robert J., 2017, Narrative economics, *American Economic Review* 107, 967–1004.
- Stein, Jeremy C, 2009, Presidential Address: Sophisticated Investors and Market Efficiency, *The Journal of Finance* 64, 1517–1548.
- U.S. House Committee on Financial Services, 2021, Game stopped? Who wins and loses when short sellers, social media, and retail investors collide, Virtual hearing, 117th Congress, First Session, February 18, 2021. Serial No. 117-3.
- Welch, Ivo, 2022, The Wisdom of the Robinhood Crowd, *The Journal of Finance* 77, 1489–1527.
- Yeung, Tai Lo, Rong Liu, Jessica A. Wachter, Michael J. Kahana, and Yongjie Zhang, 2025, Navigating through fear and greed: The experience-driven disposition effect, Working paper, available at SSRN.

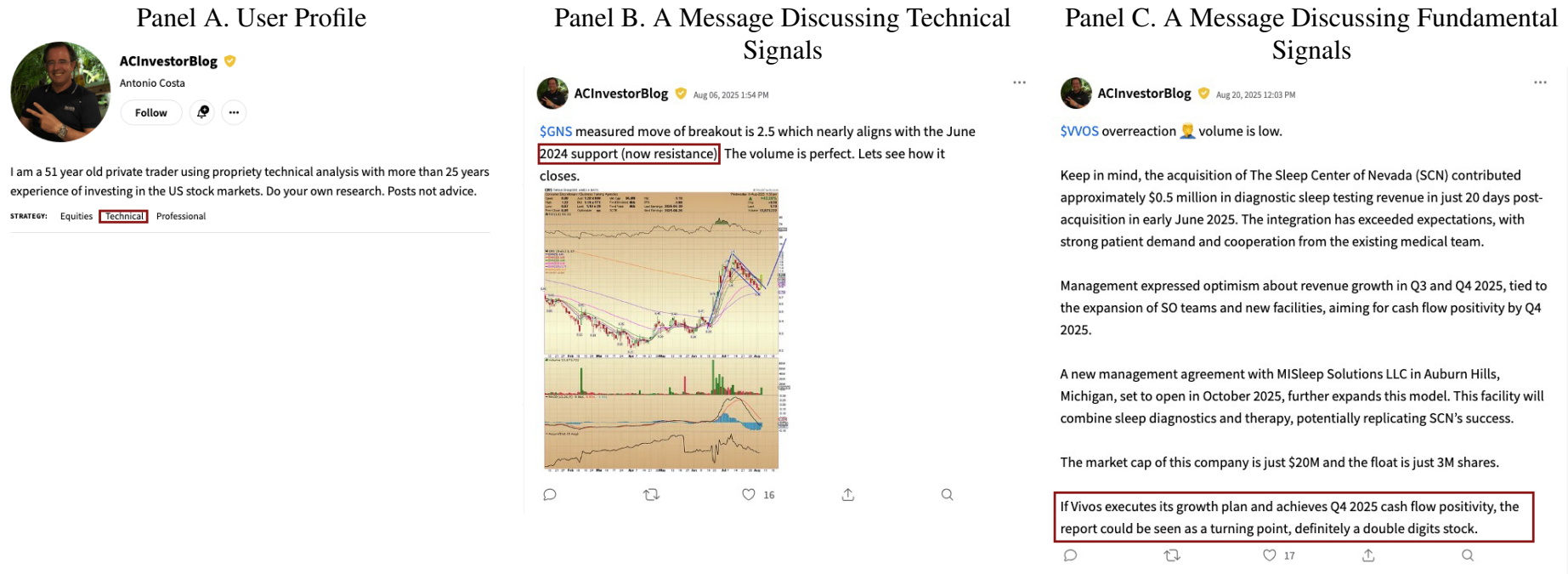


Fig. 1. Strategy Malleability: An Example

This figure presents anecdotal evidence on the malleability of retail investors' investment strategies. The screenshots are sourced from StockTwits and feature a user, ACInvestorBlog, who self-identifies as a “*Technical*” investor, as shown in the user profile in Panel A. Panel B shows a message from this user on August 6 referencing technical signals: “*\$GNS measured move of breakout is 2.5 which nearly aligns with the June 2024 support (now resistance) The volume is perfect. Lets see how it closes.*” Panel C shows a subsequent message from the same user on August 20, 2025, referencing fundamental signals: “*If Vivos executes its growth plan and achieves Q4 2025 cash flow positivity, the report could be seen as a turning point, definitely a double digits stock.*” The red rectangular boxes have been added for emphasis and are not part of the original screenshots.

[illegible][illegible]

This figure presents word clouds constructed from StockTwits messages that our LLM-based classifier assigns a TA adoption probability of at least 0.95. Panel A shows the most frequent unigrams (single words), and Panel B shows the most frequent bigrams (two-word phrases). Term sizes are proportional to their frequency within the TA message sample.

[illegible]

This figure presents word clouds constructed from StockTwits messages that our LLM-based classifier assigns an FA adoption probability of at least 0.95. Panel A shows the most frequent unigrams (single words), and Panel B shows the most frequent bigrams (two-word phrases). Term sizes are proportional to their frequency within the FA message sample.

Table 1: **Summary Statistics**

This table reports summary statistics for the full message-level sample (Panel A) and the stock-day sample (Panel B). The sample period is January 2010 to June 2023. Panel A covers all 96 million StockTwits messages in our corpus. $Usage^{TA}$ and $Usage^{FA}$ are message-level indicators equal to one if our LLM-based classifier assigns the message a Technical Analysis (TA) or Fundamental Analysis (FA) adoption probability of at least 0.95, respectively. $Usage^{OS}$ is an indicator equal to one if the message has a strategy probability above 0.95 but is classified as neither TA nor FA; these messages reference other (“OS”) strategies such as event-driven, options, or momentum trading. *Technical Investor* and *Fundamental Investor* are indicators based on the user’s self-reported investment approach (the omitted categories include *Momentum*, *Value*, *Growth*, and *Global Macro*). *Swing or Day Trader* pools users whose self-reported holding horizon is *Day Trader* or *Swing Trader*; *Long-Term Investor* flags users whose self-reported horizon is *Long Term Investor* (the omitted horizon is *Position Trader*). *Professional* and *Novice* are indicators based on the user’s self-reported experience level (the omitted level is *Intermediate*). The six self-reported indicators are set to missing for messages whose user did not provide the corresponding self-report; the reported N for these rows is therefore the number of messages from users with that self-report available, while N for the three usage indicators equals the full message count. Panel B reports statistics for the stock-day sample. $Sentiment^{TA}$ ($Sentiment^{FA}$) is net bullish sentiment among TA (FA) messages for stock i on day t . $Attention$ is the share of total StockTwits messages mentioning stock i on day t . RH_Herd indicates whether stock i is among the top ten stocks by daily percentage increase in Robinhood holders; values are in percentage points. OIB^{BJZZ} and OIB^{BHJOS} are net retail order imbalances following [Boehmer et al. \(2021\)](#) and [Barber et al. \(2023a\)](#), respectively. Firm characteristics include $Log(\text{Market Cap})$, *Book-to-Market*, *Asset Growth*, *Gross Profit-to-Assets*, *Analyst Coverage*, *Institutional Ownership*, *MAX*, *Abnormal Turnover*, and *Abnormal News Volume*. All continuous variables are winsorized at the 1% and 99% levels. For each variable, the table reports the number of non-missing observations (N), the mean, standard deviation, and the median.

Panel A: Message-Level Sample				
	N	Mean	StdDev	Median
$Usage^{TA}$	96,095,345	0.09	0.28	0
$Usage^{FA}$	96,095,345	0.14	0.35	0
$Usage^{OS}$	96,095,345	0.09	0.28	0
Technical Investor	32,144,210	0.34	0.47	0
Fundamental Investor	32,144,210	0.13	0.34	0
Swing or Day Trader	30,856,727	0.59	0.49	1
Long-Term Investor	30,856,727	0.23	0.42	0
Professional	32,355,422	0.30	0.46	0
Novice	32,355,422	0.16	0.37	0

Panel B: Stock-Level Sample				
	N	Mean	StdDev	Median
Sentiment ^{TA}	2,974,934	0.23	0.53	0.00
Sentiment ^{FA}	2,974,934	0.29	0.57	0.00
Attention	2,974,934	0.09	0.62	0.01
RH_Herd (%)	554,877	0.36	5.97	0.00
OIB ^{BJZZ}	2,974,934	-0.01	0.27	-0.00
OIB ^{BHJOS}	2,974,934	-0.01	0.25	0.00
Log(Market Cap)	2,974,934	7.12	2.52	7.14
Book-to-Market	2,974,934	0.63	0.82	0.39
Asset Growth	2,974,934	1.07	0.54	1.00
Gross Profit-to-Assets	2,974,934	0.05	0.09	0.05
Analyst Coverage	2,974,934	9.74	8.98	7.00
Institutional Ownership	2,974,934	0.60	0.32	0.69
MAX	2,974,934	0.08	0.09	0.06
Abnormal Turnover	2,974,934	-0.11	0.63	-0.11
Abnormal News Volume	2,974,934	-0.50	0.98	-0.53

Table 2: **Strategy Choice, User Attributes, and News**

This table reports message-level panel regressions of LLM-classified strategy choice on self-reported user attributes and firm-specific news. The sample period is January 2010 to June 2023. The dependent variable $Usage_{j,i,t,n}^s$ equals one if message n , posted by user j about stock i on day t , is classified as referencing strategy $s \in \{TA, FA\}$. Panel A regresses strategy choice on self-reported user attributes: *Technical Investor*, *Fundamental Investor*, *Swing or Day Trader*, *Long-Term Investor*, *Professional*, and *Novice* (the omitted approaches include *Momentum*, *Value*, *Growth*, and *Global Macro*; the omitted holding horizon is *Position Trader*; the omitted experience level is *Intermediate*). Columns (1) and (4) include only the approach indicators *Technical Investor* and *Fundamental Investor*; columns (2) and (5) add the remaining self-reported attributes; columns (3) and (6) further include stock and date fixed effects. Panel B regresses strategy choice on indicators for *Earnings News*, *Analyst News*, and *Business News* on stock i on day t , sourced from RavenPack News Analytics. Columns (1) and (3) control for the self-reported user attributes from Panel A and include stock and date fixed effects; columns (2) and (4) saturate the specification with user fixed effects (which absorb the time-invariant user attributes), so that the news effects are identified within-user. The dependent variable is multiplied by 100 in the regressions, so each coefficient measures the percentage-point change in $Usage$ associated with a unit change in the regressor. All standard errors are triple-clustered by date, stock, and user, with t -statistics in brackets. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A: User Attributes and Strategy Choice						
	$Usage_{j,i,t,n}^{TA}$			$Usage_{j,i,t,n}^{FA}$		
	(1)	(2)	(3)	(4)	(5)	(6)
Technical Investor	12.48*** [18.62]	10.18*** [14.70]	9.13*** [14.58]	-3.51*** [-5.68]	-2.27*** [-4.24]	-2.80*** [-5.89]
Fundamental Investor	-0.62 [-1.26]	-0.45 [-0.93]	-1.09** [-2.37]	8.58*** [7.92]	7.30*** [7.75]	6.28*** [7.60]
Swing or Day Trader		2.00*** [3.36]	1.95*** [3.59]		-2.91*** [-5.96]	-2.66*** [-5.81]
Long-Term Investor		-3.40*** [-7.56]	-2.86*** [-6.73]		4.58*** [5.93]	4.26*** [6.03]
Professional		4.18*** [5.61]	3.15*** [4.98]		3.12*** [3.96]	2.22*** [3.47]
Novice		-1.78*** [-6.20]	-1.39*** [-5.50]		-2.90*** [-8.18]	-2.53*** [-7.60]
Stock FEs	No	No	Yes	No	No	Yes
Date FEs	No	No	Yes	No	No	Yes
N	21,847,364	21,847,364	21,847,221	21,847,364	21,847,364	21,847,221
Adj. R ² (%)	3.09	3.88	5.93	1.04	1.84	4.29

Panel B: Information Environment and Strategy Choice				
	Usage ^{TA} _{j,i,t,n}		Usage ^{FA} _{j,i,t,n}	
	(1)	(2)	(3)	(4)
Earnings News _{i,t}	-3.21*** [-12.72]	-1.99*** [-15.41]	5.63*** [14.74]	4.51*** [20.98]
Analyst News _{i,t}	-2.17*** [-13.03]	-1.45*** [-17.84]	1.13*** [4.89]	1.34*** [10.82]
Business News _{i,t}	-2.30*** [-14.31]	-1.71*** [-19.58]	1.78*** [8.53]	1.63*** [13.59]
User Attributes	Yes	No	Yes	No
User FEs	No	Yes	No	Yes
Stock FEs	Yes	Yes	Yes	Yes
Date FEs	Yes	Yes	Yes	Yes
N	21,847,221	21,829,799	21,847,221	21,829,799
Adj. R ² (%)	4.83	18.85	4.15	18.00

Table 3: **Strategy Choice and Performance Feedback**

This table reports message-level panel regressions that examine how a user's strategy choice responds to recent strategy-specific performance, corresponding to the specification in Equation (3). The dependent variable $Usage_{j,i,t,n}^s$ equals one if message n , posted by user j about stock i on day t , is classified as referencing strategy $s \in \{TA, FA\}$, with $s = TA$ in columns (1)–(2) and $s = FA$ in columns (3)–(4). *High TA Performance* $_{j,t-1}$ (*Low TA Performance* $_{j,t-1}$) equals one if user j 's TA-implied strategy performance over the prior three months (excluding the most recent five trading days) is at or above the 75th percentile (at or below the 25th percentile) of the pooled cross-sectional distribution of TA performance, and zero otherwise. *High FA Performance* $_{j,t-1}$ and *Low FA Performance* $_{j,t-1}$ are defined analogously using FA-implied strategy performance. User strategy performance is constructed from the DGTW-adjusted returns (Daniel et al., 1997) following a user's strategy-classified messages over the prior three months. All specifications control for public-news indicators (*Earnings News*, *Analyst News*, *Business News*). Columns (2) and (4) additionally control for the user's recent intensity of TA and FA usage over the prior three months. All specifications include user, stock, and date fixed effects. The sample is restricted to users who posted at least one strategy-related (TA or FA) message in the prior three months, which reduces the sample size relative to Table 2. The dependent variable is multiplied by 100 in the regressions, so each coefficient measures the percentage-point change in *Usage* associated with a unit change in the regressor. Standard errors are triple-clustered by date, stock, and user, with t -statistics in brackets. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	$Usage_{j,i,t,n}^{TA}$		$Usage_{j,i,t,n}^{FA}$	
	(1)	(2)	(3)	(4)
High TA Performance $_{j,t-1}$	-0.17 [-1.04]	-0.17 [-1.10]	0.13 [0.86]	0.19 [1.39]
Low TA Performance $_{j,t-1}$	-0.58*** [-2.66]	-0.42** [-2.45]	0.88** [2.48]	0.84*** [2.99]
High FA Performance $_{j,t-1}$	0.02 [0.13]	0.04 [0.31]	-0.30* [-1.67]	-0.30* [-1.85]
Low FA Performance $_{j,t-1}$	0.39** [2.45]	0.33** [2.09]	-0.68*** [-3.40]	-0.60*** [-3.23]
Past Strategy Usage	No	Yes	No	Yes
Public News	Yes	Yes	Yes	Yes
User FEs	Yes	Yes	Yes	Yes
Stock FEs	Yes	Yes	Yes	Yes
Date FEs	Yes	Yes	Yes	Yes
N	4,716,661	4,716,661	4,716,661	4,716,661
Adj. R ² (%)	9.92	9.99	10.25	10.31

Table 4: **Strategy Choice and Social Feedback**

This table reports message-level panel regressions that examine how a user's strategy choice responds to the social feedback (likes) received on recent strategy-classified messages, corresponding to the specification in Equation (4). The dependent variable $Usage_{j,i,t,n}^s$ equals one if message n , posted by user j about stock i on day t , is classified as referencing strategy $s \in \{TA, FA\}$, with $s = TA$ in columns (1)–(2) and $s = FA$ in columns (3)–(4). *High TA Likes* $_{j,t-1}$ (*Low TA Likes* $_{j,t-1}$) equals one if the total number of likes user j received on TA messages posted in the prior three months (excluding the most recent five trading days) is at or above the 75th percentile (at or below the 25th percentile) of the pooled cross-sectional distribution of TA-message likes, and zero otherwise. *High FA Likes* $_{j,t-1}$ and *Low FA Likes* $_{j,t-1}$ are defined analogously for FA messages. All specifications control for public-news indicators (*Earnings News*, *Analyst News*, *Business News*) and the user's recent intensity of TA and FA usage over the prior three months. Columns (2) and (4) additionally control for *High/Low TA Performance* $_{j,t-1}$ and *High/Low FA Performance* $_{j,t-1}$ (defined as in Table 3). All specifications include user, stock, and date fixed effects. The sample is restricted to users who posted at least one strategy-related (TA or FA) message in the prior three months. The dependent variable is multiplied by 100 in the regressions, so each coefficient measures the percentage-point change in *Usage* associated with a unit change in the regressor. Standard errors are triple-clustered by date, stock, and user, with t -statistics in brackets. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	$Usage_{j,i,t,n}^{TA}$		$Usage_{j,i,t,n}^{FA}$	
	(1)	(2)	(3)	(4)
High TA Likes $_{j,t-1}$	0.85*** [3.04]	0.84*** [3.02]	-0.87*** [-2.96]	-0.85*** [-2.88]
Low TA Likes $_{j,t-1}$	-0.83*** [-4.01]	-0.82*** [-4.05]	1.13*** [5.30]	1.11*** [5.09]
High FA Likes $_{j,t-1}$	-1.43*** [-6.16]	-1.42*** [-6.13]	1.33*** [3.54]	1.30*** [3.49]
Low FA Likes $_{j,t-1}$	1.70*** [8.27]	1.69*** [8.24]	-1.35*** [-5.95]	-1.33*** [-5.74]
Past Performance	No	Yes	No	Yes
Past Strategy Usage	Yes	Yes	Yes	Yes
Public News	Yes	Yes	Yes	Yes
User FEs	Yes	Yes	Yes	Yes
Stock FEs	Yes	Yes	Yes	Yes
Date FEs	Yes	Yes	Yes	Yes
N	4,716,661	4,716,661	4,716,661	4,716,661
Adj. R ² (%)	10.01	10.01	10.32	10.32

Table 5: **Strategy Sentiment and Next-Day Returns**

This table reports stock-day predictive regressions of next-day returns on strategy-specific sentiment over January 2010 to June 2023, corresponding to the specification in Eq. (5). The dependent variable $Return_{i,t+1}$ is the next-day return on stock i . $Sentiment_{i,t}^s$ is the stock-day sentiment for strategy $s \in \{TA, FA\}$, computed as the difference between the bullish and bearish message counts in category s , scaled by the total number of messages in that category. Columns (1) and (2) include TA and FA sentiment separately; column (3) includes both jointly. $Attention_{i,t}$ is the share of total StockTwits messages posted on day t that reference stock i . The control vector $X_{i,t}$ comprises $Log(\text{Market Cap})$, $Log(\text{Book-to-Market})$, $Asset\ Growth$, $Gross\ Profit\text{-}to\text{-}Assets$, $Log(1+Analyst\ Coverage)$, $Log(\text{Institutional Ownership})$, MAX , $Abnormal\ Turnover$, $Abnormal\ News\ Volume$, and the five most recent daily return lags. All specifications include date fixed effects. The dependent variable is multiplied by 10,000 in the regressions, so each coefficient measures the change in next-day return, in basis points, associated with a unit change in the regressor. Standard errors are clustered by trading day, with t -statistics in brackets. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Return $_{i,t+1}$		
	(1)	(2)	(3)
Sentiment $_{i,t}^{TA}$	-1.61** [-2.19]		-1.81** [-2.52]
Sentiment $_{i,t}^{FA}$		1.37** [2.17]	1.57** [2.56]
Attention $_{i,t}$	-5.58*** [-5.47]	-5.61*** [-5.50]	-5.60*** [-5.49]
Stock Characteristics	Yes	Yes	Yes
Lagged Returns	Yes	Yes	Yes
Date FEs	Yes	Yes	Yes
N	2,974,304	2,974,304	2,974,304
Adj. R ² (%)	8.76	8.76	8.76

Table 6: Strategy Sentiment and Long-Short Returns

This table reports the performance of daily long-short (L/S) portfolios formed on StockTwits strategy-specific sentiment. The sample period is January 2010 to June 2023. Stock-day sentiment for strategy $s \in \{TA, FA\}$ is the difference between bullish and bearish message counts in that category, scaled by the total number of messages in that category. To mitigate confounding effects from known return predictors, on each day, we cross-sectionally regress the strategy-specific sentiment on stock characteristics and retain the residual sentiment. Following [Jensen et al. \(2023\)](#), the L/S portfolio takes weights proportional to each stock's residual sentiment deviation from the daily cross-sectional mean. The TA-based L/S portfolio multiplies the residual TA sentiment by -1 (i.e., long low-TA-sentiment stocks, short high-TA-sentiment stocks). Panel A reports annualized average daily raw returns (in percent) along with the Sharpe ratio (SR); Panel B reports annualized DGTW-adjusted abnormal returns ([Daniel et al., 1997](#)) along with the information ratio (IR). t -statistics for the average returns are reported in column (2).

Panel A. Raw Return			
	Mean (annual, %)	t -statistic	SR (annual)
	(1)	(2)	(3)
TA	9.50	2.91	0.86
FA	6.48	2.04	0.58

Panel B. DGTW Excess Return			
	Mean (annual, %)	t -statistic	IR (annual)
	(1)	(2)	(3)
TA	10.10	3.21	1.00
FA	7.75	2.53	0.75

Table 7: Strategy Sentiment and Longer-Horizon Returns

This table extends the predictive regression in Table 5 (Eq. (5)) to longer forecasting horizons. The dependent variable is the cumulative stock return over horizon h , with h equal to $t + 2$ to $t + 5$ in column (1), $t + 6$ to $t + 10$ in column (2), and $t + 11$ to $t + 15$ in column (3). The right-hand-side variables, control vector $X_{i,t}$, and fixed effects are as in Table 5. The dependent variable is multiplied by 10,000 in the regressions, so each coefficient measures the change in cumulative return, in basis points, associated with a unit change in the regressor. Standard errors are clustered by trading day, with t -statistics in brackets. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Return $_{i,t+2 \rightarrow t+5}$	Return $_{i,t+6 \rightarrow t+10}$	Return $_{i,t+11 \rightarrow t+15}$
	(1)	(2)	(3)
Sentiment $_{i,t}^{TA}$	-5.36*** [-3.66]	-6.07*** [-3.76]	-1.94 [-1.23]
Sentiment $_{i,t}^{FA}$	2.30* [1.88]	-0.44 [-0.32]	-0.28 [-0.21]
Attention $_{i,t}$	-10.43*** [-6.89]	-8.35*** [-6.08]	-6.54*** [-5.37]
Stock Characteristics	Yes	Yes	Yes
Lagged Returns	Yes	Yes	Yes
Date FEs	Yes	Yes	Yes
N	2,972,337	2,970,213	2,968,174
Adj. R ² (%)	10.68	10.67	11.14

Table 8: **Social Feedback and Sentiment Informativeness**

This table examines how social feedback (likes) modulates the return predictability of strategy-specific sentiment, corresponding to the specification in Eq. (6). Stock-day sentiment is interacted with the log of one plus the total number of likes received by strategy-classified messages on stock i on day t . The dependent variable $Return_{i,h}$ is the compounded cumulative return over horizon h : $h = t + 1$ in column (1), $t + 2$ to $t + 5$ in column (2), $t + 6$ to $t + 10$ in column (3), and $t + 11$ to $t + 15$ in column (4). Other variables and controls are defined as in Table 5. All specifications include date fixed effects. The dependent variable is multiplied by 10,000 in the regressions, so each coefficient measures the change in cumulative return, in basis points, associated with a unit change in the regressor. Standard errors are clustered by trading day, with t -statistics in brackets. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	$Return_{i,t+1}$	$Return_{i,t+2 \rightarrow t+5}$	$Return_{i,t+6 \rightarrow t+10}$	$Return_{i,t+11 \rightarrow t+15}$
	(1)	(2)	(3)	(4)
$Sentiment_{i,t}^{TA} \times \text{Log}(1+Likes_{i,t}^{TA})$	-5.90*** [-5.25]	-15.78*** [-6.82]	-12.48*** [-4.85]	-3.21 [-1.24]
$Sentiment_{i,t}^{FA} \times \text{Log}(1+Likes_{i,t}^{FA})$	2.62*** [2.60]	6.83*** [3.45]	6.83*** [3.25]	-0.95 [-0.45]
$Sentiment_{i,t}^{TA}$	2.30*** [3.22]	3.74*** [2.64]	0.16 [0.10]	1.10 [0.75]
$Sentiment_{i,t}^{FA}$	1.60** [2.38]	1.36 [1.02]	-0.63 [-0.43]	2.01 [1.38]
$\text{Log}(1+Likes_{i,t}^{TA})$	-0.59 [-0.63]	1.51 [0.82]	5.58*** [2.76]	0.15 [0.08]
$\text{Log}(1+Likes_{i,t}^{FA})$	-2.46** [-2.54]	-4.80** [-2.57]	-8.66*** [-4.51]	-2.97 [-1.55]
$Attention_{i,t}$	-3.77*** [-3.63]	-7.69*** [-5.05]	-5.84*** [-4.10]	-4.62*** [-3.50]
Stock Characteristics	Yes	Yes	Yes	Yes
Lagged Returns	Yes	Yes	Yes	Yes
Date FEs	Yes	Yes	Yes	Yes
N	2,974,304	2,972,337	2,970,213	2,968,174
Adj. R ² (%)	8.76	10.68	10.68	11.14

Table 9: **Strategy Sentiment and Intense Robinhood Buying**

This table examines the predictive relation between strategy-specific sentiment, interacted with social feedback, and subsequent intense buying episodes by Robinhood investors, corresponding to the specification in Eq. (7). The dependent variable $RH_Herd_{i,t+h}$ is an indicator equal to one if stock i is among the top ten stocks ranked by the daily percentage increase in the number of Robinhood users holding stock i on day $t+h$, conditional on at least 100 users holding the stock at the end of day $t+h-1$ (Barber et al., 2022). Columns (1)–(3) report results for $h = 1, 2, 3$. Robinhood user holdings data, from RobinTrack, span May 2018 to August 2020. Sentiment variables, *Attention*, and the firm-level controls $X_{i,t}$ are defined as in Table 5. All specifications include date fixed effects. The dependent variable is multiplied by 10,000 in the regressions, so each coefficient measures the basis-point change in the probability of an intense-buying episode associated with a unit change in the regressor. Standard errors are clustered by stock and trading day, with t -statistics in brackets. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	RH_Herd _{<i>i,t+1</i>}	RH_Herd _{<i>i,t+2</i>}	RH_Herd _{<i>i,t+3</i>}
	(1)	(2)	(3)
Sentiment _{<i>i,t</i>} ^{TA} × Log(1+Likes _{<i>i,t</i>} ^{TA})	7.44*** [3.48]	4.31** [2.22]	1.30 [0.83]
Sentiment _{<i>i,t</i>} ^{FA} × Log(1+Likes _{<i>i,t</i>} ^{FA})	-4.77** [-2.56]	-3.82*** [-2.59]	1.13 [0.88]
Sentiment _{<i>i,t</i>} ^{TA}	-0.71 [-0.44]	-0.23 [-0.14]	2.19 [1.49]
Sentiment _{<i>i,t</i>} ^{FA}	4.35*** [2.90]	4.41*** [2.99]	-0.20 [-0.16]
Log(1+Likes _{<i>i,t</i>} ^{TA})	-3.05* [-1.72]	-1.65 [-1.13]	-1.47 [-1.22]
Log(1+Likes _{<i>i,t</i>} ^{FA})	2.82* [1.81]	0.50 [0.42]	-1.17 [-1.11]
RH_Herd _{<i>i,t</i>}	0.10*** [15.01]	0.02*** [4.97]	0.01*** [3.22]
Attention _{<i>i,t</i>}	14.17** [2.46]	6.45* [1.65]	4.93* [1.71]
Stock Characteristics	Yes	Yes	Yes
Lagged Returns	Yes	Yes	Yes
Date FEs	Yes	Yes	Yes
N	541,977	538,955	533,567
Adj. R ² (%)	1.63	0.15	0.10

Table 10: **Strategy Sentiment and Short Squeezes**

This table examines the predictive relation between strategy-specific sentiment, interacted with social feedback, and subsequent short-squeeze episodes. The dependent variable $Squeeze_{i,t+h}$ is an indicator equal to one if stock i experiences a short-squeeze event on day $t + h$, defined as a stock-day in which the one-day stock return exceeds 30% but the daily percentage change in short loan value decreases more than 20%. Columns (1)–(3) report results for $h = 1, 2, 3$. Sentiment variables, *Attention*, and the firm-level controls $X_{i,t}$ are defined as in Table 5. All specifications include date fixed effects. The dependent variable is multiplied by 10,000 in the regressions, so each coefficient measures the basis-point change in the probability of a short-squeeze event associated with a unit change in the regressor. Standard errors are clustered by stock and trading day, with t -statistics in brackets. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Squeeze _{<i>i,t+1</i>}	Squeeze _{<i>i,t+2</i>}	Squeeze _{<i>i,t+3</i>}
	(1)	(2)	(3)
Sentiment _{<i>i,t</i>} ^{TA} × Log(1+Likes _{<i>i,t</i>} ^{TA})	1.03*** [2.84]	0.88** [2.46]	0.68** [2.20]
Sentiment _{<i>i,t</i>} ^{FA} × Log(1+Likes _{<i>i,t</i>} ^{FA})	-0.42 [-1.59]	-0.41 [-1.50]	0.04 [0.16]
Sentiment _{<i>i,t</i>} ^{TA}	-0.37* [-1.75]	-0.42** [-2.20]	-0.37* [-1.93]
Sentiment _{<i>i,t</i>} ^{FA}	-0.01 [-0.03]	0.09 [0.57]	0.03 [0.22]
Log(1+Likes _{<i>i,t</i>} ^{TA})	0.62** [2.13]	0.70** [2.49]	0.65** [2.57]
Log(1+Likes _{<i>i,t</i>} ^{FA})	-0.02 [-0.07]	-0.03 [-0.12]	-0.20 [-1.06]
Attention _{<i>i,t</i>}	2.14* [1.72]	1.31 [1.62]	1.20 [1.64]
Stock Characteristics	Yes	Yes	Yes
Lagged Returns	Yes	Yes	Yes
Date FEs	Yes	Yes	Yes
N	2,957,923	2,957,537	2,957,052
Adj. R ² (%)	0.11	0.08	0.07

Table 11: **Strategy Sentiment and Aggregate Retail Order Flow**

This table examines the contemporaneous relation between StockTwits strategy-specific sentiment and aggregate retail order imbalance over January 2010 to June 2023, corresponding to the specification in Eq. (10). The dependent variable $OIB_{i,t}$ is the aggregate net retail order imbalance on stock i on day t , defined as marketable retail buy dollar volume minus marketable retail sell dollar volume, scaled by the sum of the two. Column (1) uses the BJZZ identification algorithm of [Boehmer et al. \(2021\)](#); column (2) uses the BHJOS algorithm of [Barber et al. \(2023a\)](#). To align the sentiment measures with intraday retail orders, strategy-specific sentiment is computed using only messages posted during regular trading hours (9:30–16:00 ET). Sentiment variables, $Attention_{i,t}$, and the firm-level controls $X_{i,t}$ are defined as in Table 5. All specifications include date fixed effects. The dependent variable is multiplied by 100 in the regressions, so each coefficient measures the percentage-point change in net retail buying associated with a unit change in the regressor. Standard errors are clustered by stock and trading day, with t -statistics in brackets. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	$OIB_{i,t}^{BJZZ}$	$OIB_{i,t}^{BHJOS}$
	(1)	(2)
Sentiment $_{i,t}^{TA}$	0.87*** [19.58]	1.14*** [27.39]
Sentiment $_{i,t}^{FA}$	0.67*** [17.29]	0.74*** [19.05]
Attention $_{i,t}$	0.39** [2.57]	0.49*** [3.13]
Stock Characteristics	Yes	Yes
Lagged Returns	Yes	Yes
Date FEs	Yes	Yes
N	2,974,934	2,974,934
Adj. R ² (%)	0.72	1.02

Table 12: **Strategy-Linked Retail Order Imbalance and Returns**

This table examines whether the strategy-driven components of aggregate retail order imbalance (OIB) carry differential information about future stock returns. For each retail-order identification algorithm $j \in \{BJZZ, BHJOS\}$, we decompose $OIB_{i,t}^j$ into a TA-driven component, an FA-driven component, and a residual using the fitted values from the contemporaneous regression in Eq. (10): $OIB_{i,t}^{j,s} \equiv \hat{\beta}_1^s \times \text{Sentiment}_{i,t}^s$ for $s \in \{TA, FA\}$, with the residual $OIB_{i,t}^{j,Resid}$ capturing the portion of OIB unrelated to TA and FA sentiment. To align the sentiment measures with intraday retail orders, strategy-specific sentiment in Eq. (10) is constructed using only messages posted during regular trading hours (9:30–16:00 ET). We then re-estimate the predictive regression in Eq. (5), replacing strategy-specific sentiment with the three decomposed OIB components ($OIB_{i,t}^{j,TA}$, $OIB_{i,t}^{j,FA}$, and $OIB_{i,t}^{j,Resid}$). The dependent variable is the cumulative stock return over horizon h , with $h = t + 1$ in column (1), $t + 2$ to $t + 5$ in column (2), $t + 6$ to $t + 10$ in column (3), and $t + 11$ to $t + 15$ in column (4). Panel A uses the BJZZ identification algorithm of [Boehmer et al. \(2021\)](#); Panel B uses the BHJOS algorithm of [Barber et al. \(2023a\)](#). The control variables are the same as in Table 5. All specifications include trading day fixed effects. The dependent variable is multiplied by 10,000 in the regressions, so each coefficient measures the change in cumulative return, in basis points, associated with a unit change in the regressor. Standard errors are clustered by trading day, with corresponding t -statistics in brackets. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A: BJZZ				
	Return _{$i,t+1$}	Return _{$i,t+2 \rightarrow t+5$}	Return _{$i,t+6 \rightarrow t+10$}	Return _{$i,t+11 \rightarrow t+15$}
	(1)	(2)	(3)	(4)
$OIB_{i,t}^{BJZZ,TA}$	-3.12** [-2.52]	-9.23*** [-3.66]	-10.47*** [-3.76]	-3.35 [-1.23]
$OIB_{i,t}^{BJZZ,FA}$	3.90** [2.56]	5.70* [1.88]	-1.08 [-0.32]	-0.71 [-0.21]
$OIB_{i,t}^{BJZZ,Resid}$	0.17*** [13.52]	0.08*** [3.45]	0.08*** [2.94]	0.03 [1.01]
Attention _{i,t}	-5.60*** [-5.49]	-10.43*** [-6.89]	-8.35*** [-6.08]	-6.54*** [-5.37]
Stock Characteristics	Yes	Yes	Yes	Yes
Lagged Returns	Yes	Yes	Yes	Yes
Date FEs	Yes	Yes	Yes	Yes
N	2,974,304	2,972,337	2,970,213	2,968,174
Adj. R ² (%)	8.76	10.68	10.67	11.14

Panel B: BHJOS				
	Return _{<i>i,t</i>+1}	Return _{<i>i,t</i>+2→<i>t</i>+5}	Return _{<i>i,t</i>+6→<i>t</i>+10}	Return _{<i>i,t</i>+11→<i>t</i>+15}
	(1)	(2)	(3)	(4)
OIB _{<i>i,t</i>} ^{BHJOS,TA}	-2.01** [-2.52]	-5.95*** [-3.66]	-6.75*** [-3.76]	-2.16 [-1.23]
OIB _{<i>i,t</i>} ^{BHJOS,FA}	3.01** [2.56]	4.39* [1.88]	-0.83 [-0.32]	-0.54 [-0.21]
OIB _{<i>i,t</i>} ^{BHJOS,Resid}	0.24*** [18.12]	0.18*** [6.72]	0.08*** [2.61]	0.05 [1.52]
Attention _{<i>i,t</i>}	-5.60*** [-5.49]	-10.43*** [-6.89]	-8.35*** [-6.08]	-6.54*** [-5.37]
Stock Characteristics	Yes	Yes	Yes	Yes
Lagged Returns	Yes	Yes	Yes	Yes
Date FEs	Yes	Yes	Yes	Yes
N	2,974,304	2,972,337	2,970,213	2,968,174
Adj. R ² (%)	8.77	10.68	10.67	11.14

Appendix A.

A.1. Comparing Classification Approaches

Our main approach is to first generate message-level strategy labels using GPT-4 Turbo and then use these labels to fine-tune a BERT model (TA-BERT) for our classification task. We compare this method against two alternative approaches: (i) fine-tuning a BERT model on users’ self-declared investment approaches and (ii) a traditional dictionary-based method.

We begin by characterizing the predictions of our TA-BERT and FA-BERT models. Figures A.1 and A.2 plot the distribution of *TA Adoption Probability* across investor groups defined by self-declared investment approaches and horizons, respectively. A key property of TA-BERT is that its predictions are strongly bimodal: most messages receive either a very low probability score ($< 5\%$) or a very high probability score ($> 95\%$), indicating that the model classifies the vast majority of messages unambiguously. The predictions are also positively correlated with users’ self-declared profiles—users who identify as technical or momentum traders post a substantially higher share of high-probability TA messages than those identifying as fundamental, value, growth, or global macro investors, while users with shorter investment horizons (e.g., day traders) rely more heavily on TA than long-term investors. Figures A.3 and A.4 document the analogous properties for FA-BERT: the model again produces a bimodal distribution, and the fraction of high-probability FA messages is highest among users who self-report fundamental, value, and growth investment approaches.

We evaluate the alternative model trained on self-declared labels. Since some StockTwits users self-declare their investment approach, we fine-tune a BERT model on messages from these users. To do this, we randomly sample 20,000 messages from this group and assign a message-level label based on the user’s self-declaration. After fine-tuning, we apply this model to classify all StockTwits messages. Figure A.5 presents the resulting predicted probabilities. The model fails to produce a bimodal distribution and therefore fails to reliably classify strategies at the message level. For example, the top-left panel shows that for messages from self-declared technical investors, the predicted TA probability is relatively low, with most messages scoring below 50%. This poor performance likely reflects the fact that investors use diverse

strategies and often post messages that do not align with their primary self-declared approach.

Next, we compare our TA-BERT model against the self-declared model using a manually labeled benchmark. Figure A.6 displays the TA probability for 500 human-classified messages. The top panels show that TA-BERT produces a clear, bimodal separation: most human-labeled technical messages receive a high TA probability, while most non-technical messages receive a low score. In contrast, the bottom panels show that the BERT model trained on self-declared approaches fails to clearly differentiate between the two message types. While the average TA probability is higher for human-labeled technical messages, the distribution is not bimodal, making it difficult to set a clear classification threshold.

Finally, we assess the dictionary-based bag-of-words (BoW) approach. Figures A.7 and A.8 present the TA and FA intensity scores calculated using the word lists from Cookson and Niessner (2020). Similar to the self-declared model, the BoW approach does not produce a bimodal distribution. For instance, messages from self-declared technical investors largely receive a BoW TA intensity score near zero, and messages from most fundamental investors receive a BoW FA score near zero.

Taken together, our analyses highlight the effectiveness of large language models in generating high-quality training data and demonstrate the ability of smaller, fine-tuned models to learn efficiently from such data for specific classification tasks.

A.2. Additional Tables and Figures

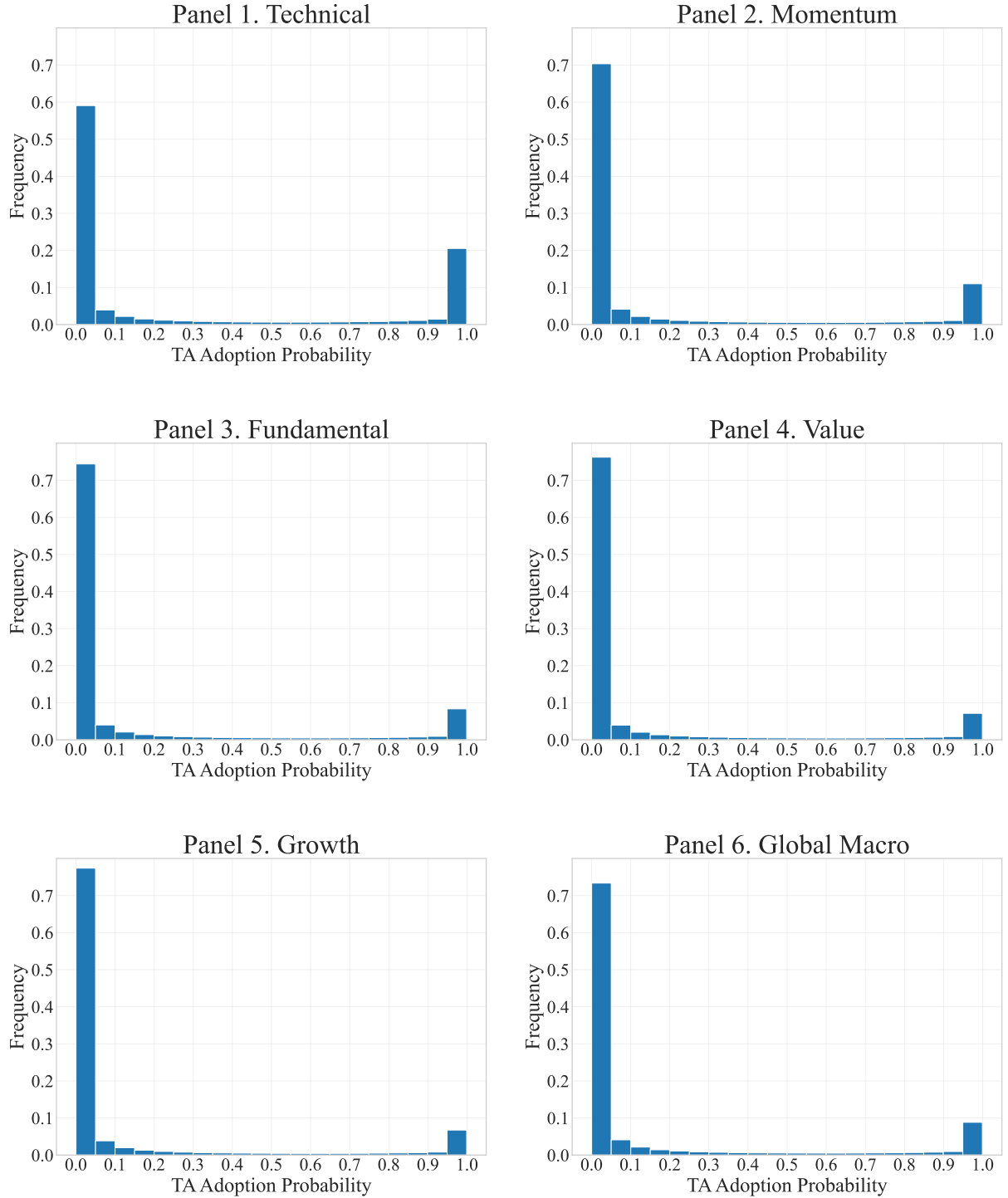


Fig. A.1. TA Classification by Self-Declared Investment Approach

This figure plots the distribution of *Technical Analysis (TA) Adoption Probability* over Stock-Twits messages, grouped by users' self-declared investment approaches. Each message receives a probabilistic score from our fine-tuned *TA-BERT* model, where higher values indicate a greater likelihood that the message employs technical analysis. For each investment approach group, the histogram reports the frequency of messages across levels of *TA Adoption Probability*.

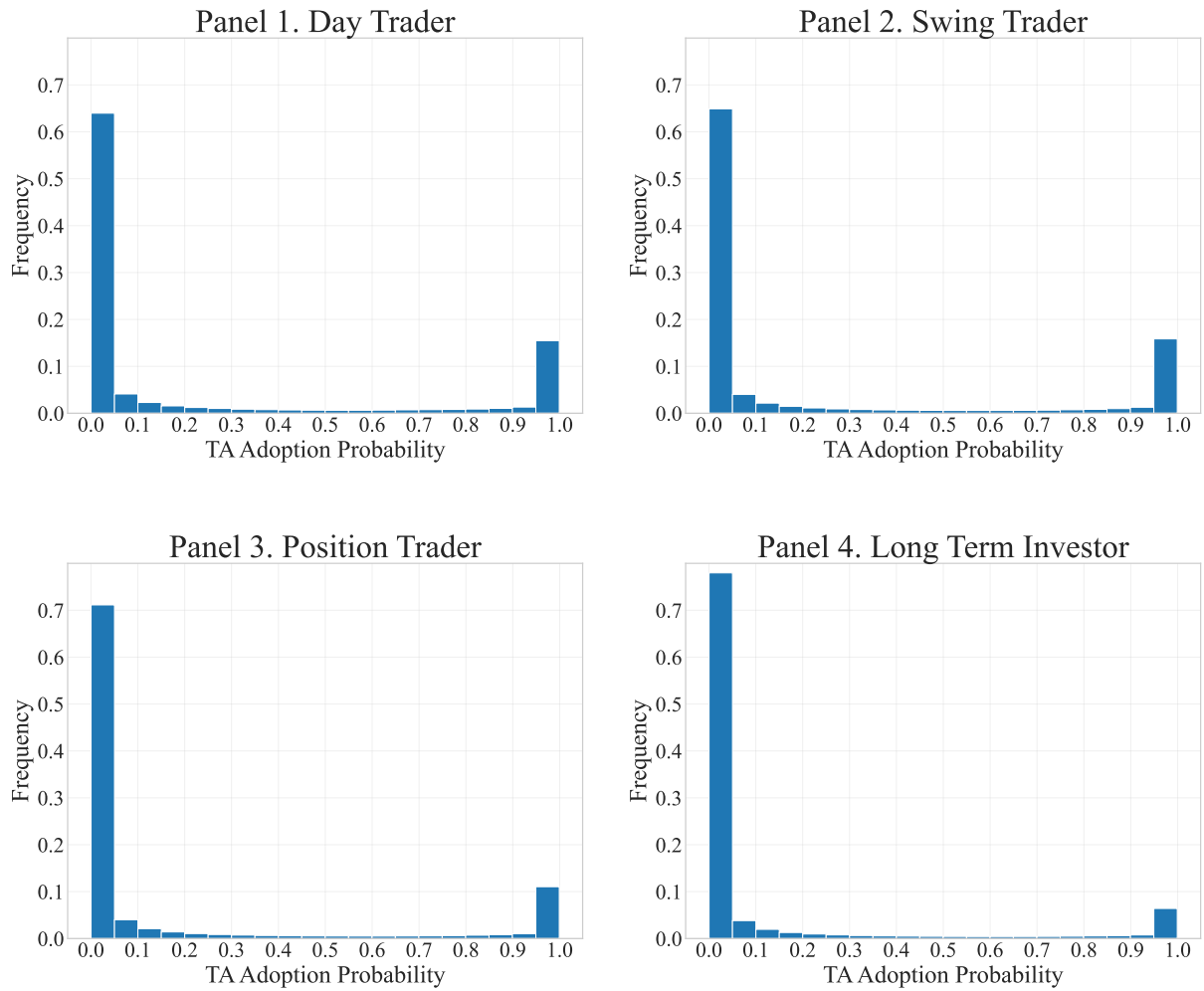


Fig. A.2. TA Classification by Self-Declared Investment Horizon

This figure plots the distribution of *Technical Analysis (TA) Adoption Probability* over Stock-Twits messages, grouped by users' self-declared investment horizons. Each message receives a probabilistic score from our fine-tuned *TA-BERT* model, where higher values indicate a greater likelihood that the message employs technical analysis. For each investment horizon group, the histogram reports the frequency of messages across levels of *TA Adoption Probability*.

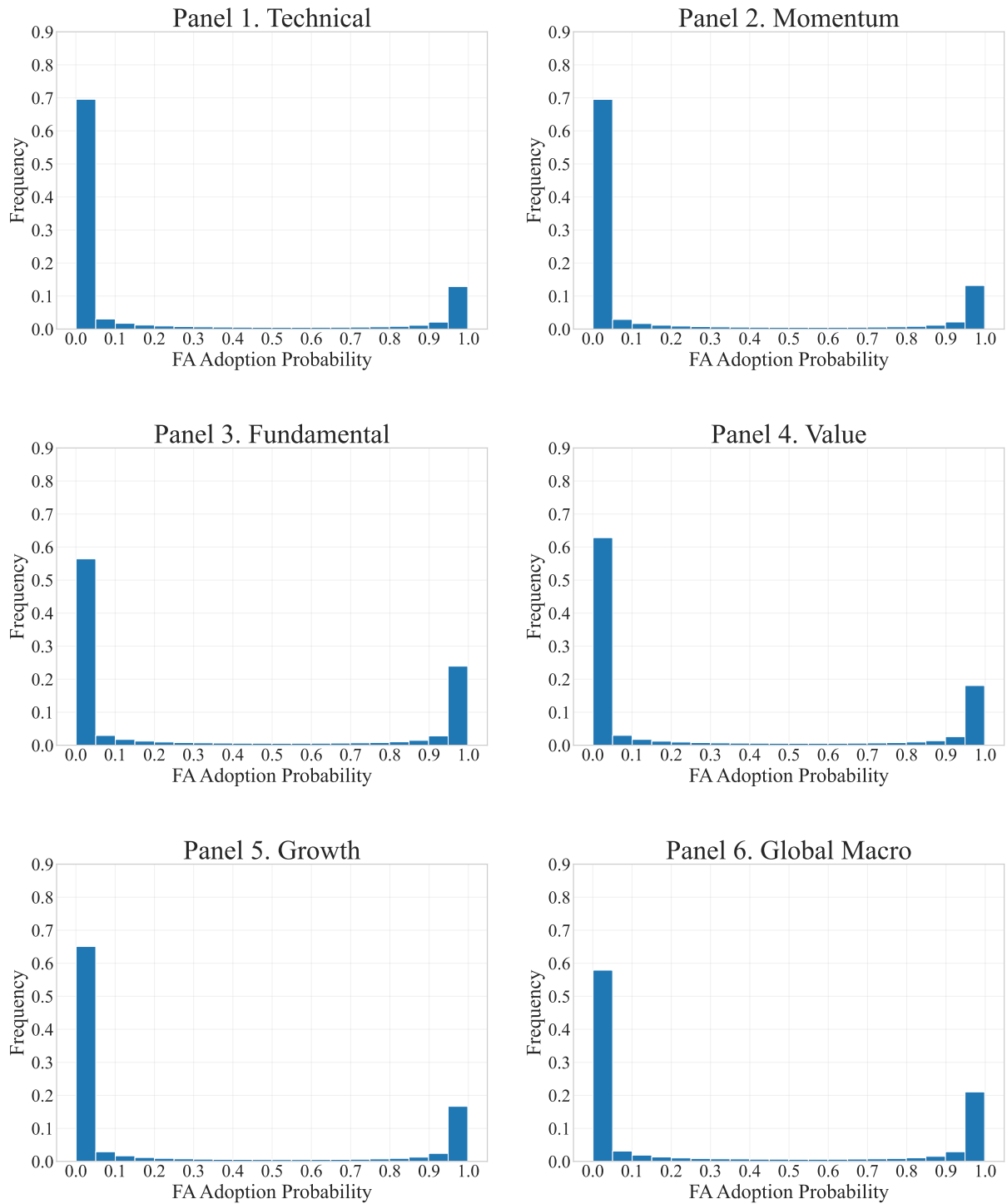


Fig. A.3. FA Classification by Self-Declared Investment Approach

This figure shows the distribution of *Fundamental Analysis (FA) Adoption Probability* over StockTwits messages, grouped by users' self-declared investment approaches. Each message receives a probabilistic score from our fine-tuned *FA-BERT* model, where higher values indicate a greater likelihood that the message employs fundamental analysis. For each investment approach group, the histogram reports the frequency of messages across levels of *FA Adoption Probability*.

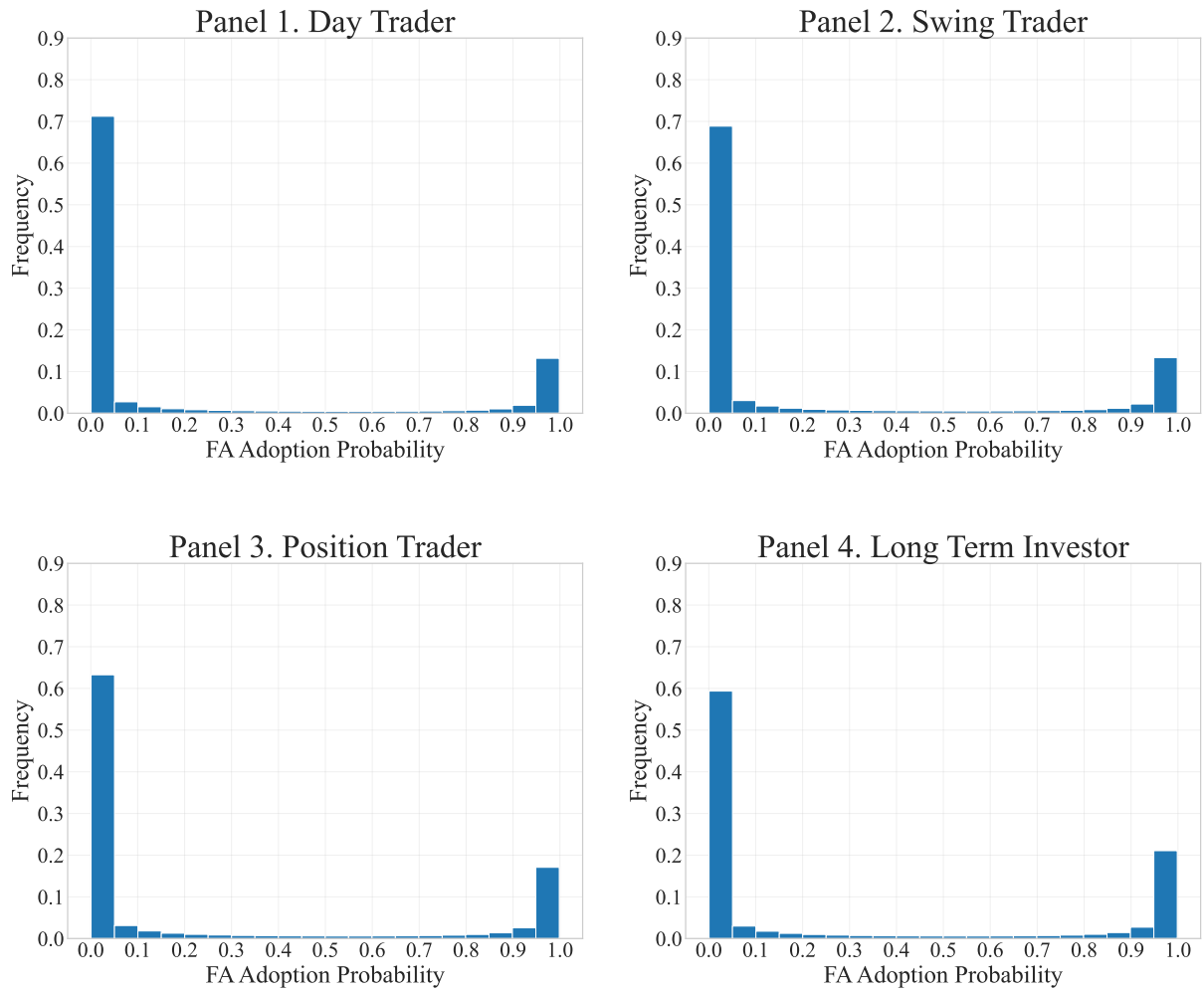


Fig. A.4. FA Classification by Self-Declared Investment Horizon

This figure presents the distribution of *Fundamental Analysis (FA) Adoption Probability* over StockTwits messages, grouped by users' self-declared investment horizons. Each message receives a probabilistic score from our fine-tuned FA-BERT model, where higher values indicate a greater likelihood that the message employs fundamental analysis. For each investment horizon group, the histogram reports the frequency of messages across levels of *FA Adoption Probability*.

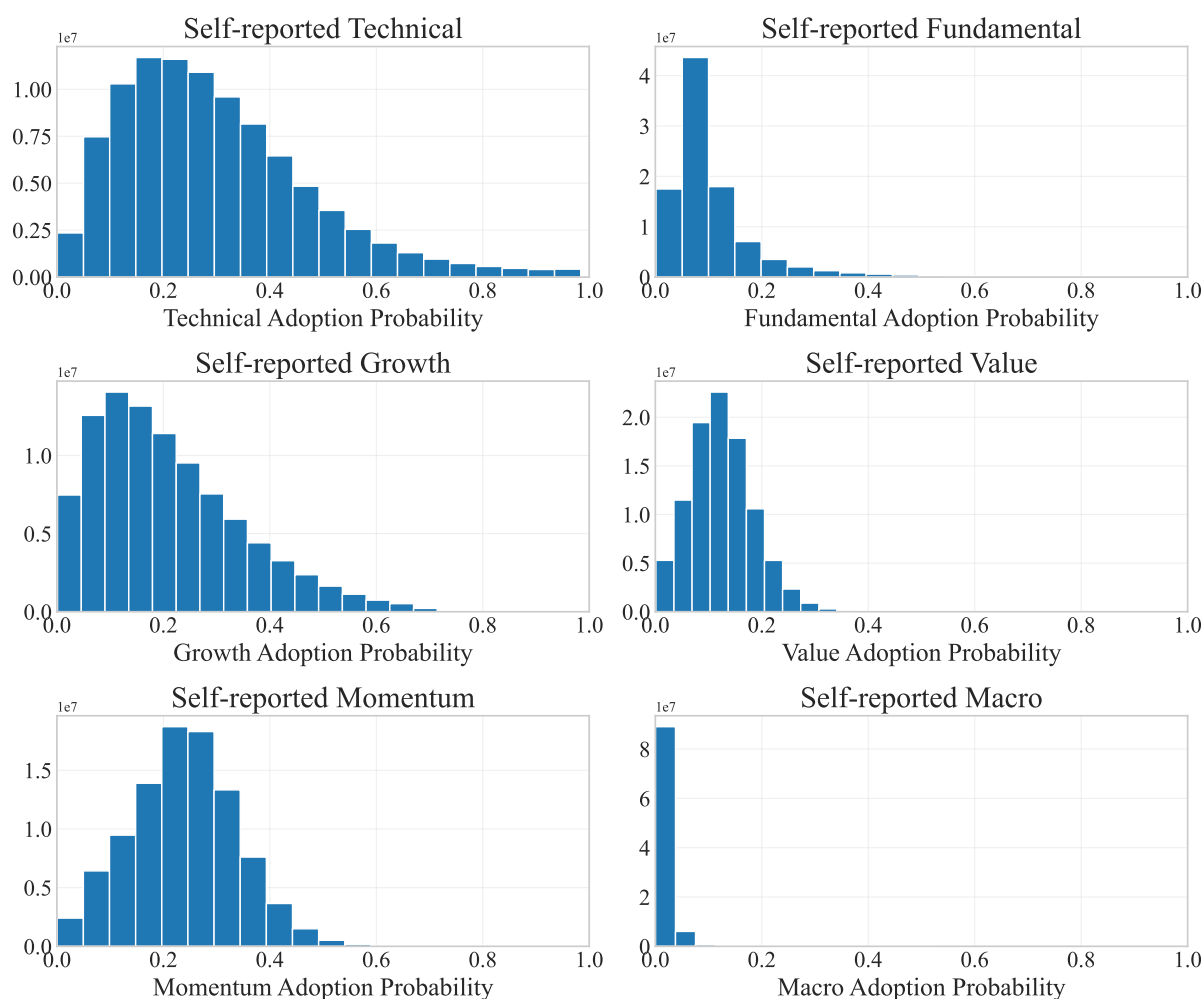


Fig. A.5. Classification Using Self-Declared Approaches

This figure presents the distribution of investment approach classification probabilities from a BERT model fine-tuned on a random sample of messages combined with users' self-declared investment approaches. The model assigns each message a probability of belonging to one of six distinct investment approaches. For each self-declared approach, the histogram reports the frequency distribution of messages across levels of BERT-classified probabilities that a message is identified as belonging to the same approach.

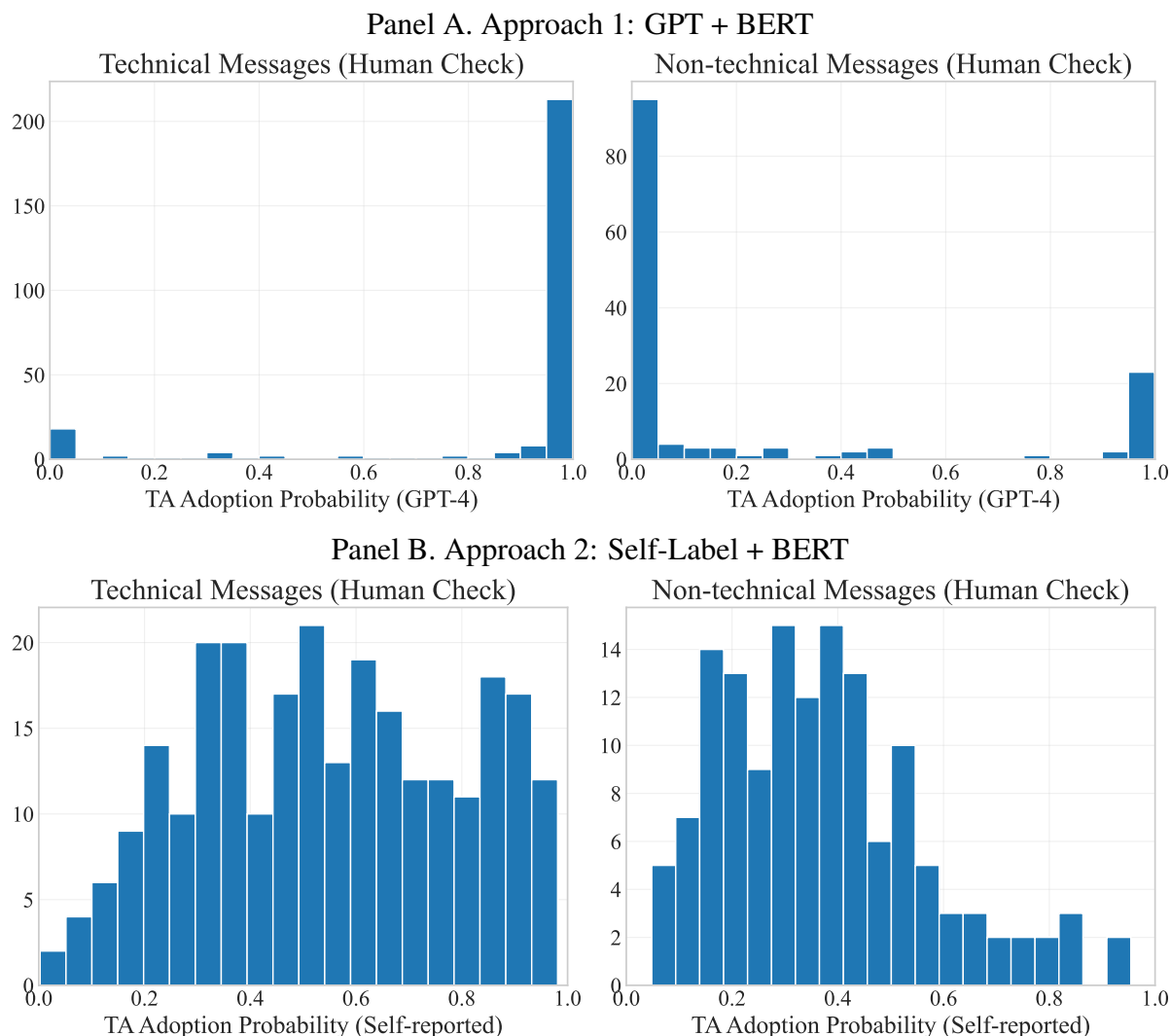


Fig. A.6. Strategy Classification: Validation Against Human Labels

This figure compares the performance of two alternative strategy-classification methods. The first approach uses GPT-4-Turbo to classify a random sample of messages, and the resulting labels are used to fine-tune a BERT model. The second approach fine-tunes a BERT model using a random sample of messages with users' self-declared investment approaches. To evaluate performance, we construct a ground truth benchmark consisting of two message samples classified by research assistants: (I) Human-labeled technical messages and (II) Human-labeled non-technical messages. Each method is then applied to classify the benchmark messages and produces probabilities that each message relies on technical analysis. Panel A presents histograms of technical-analysis probabilities from the first approach for Sample I (left) and Sample II (right). Panel B presents the corresponding histograms from the second approach.

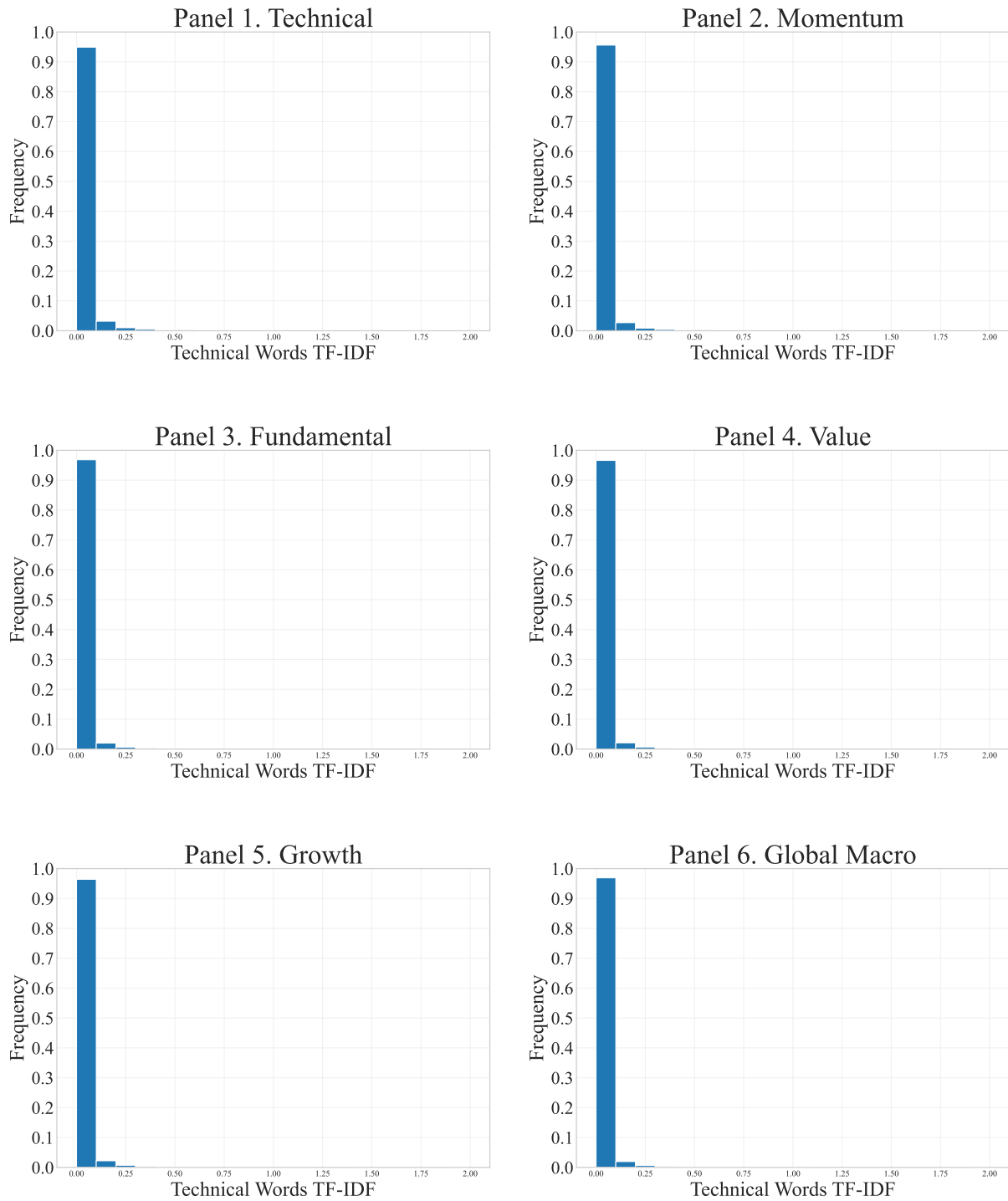


Fig. A.7. Dictionary-Based TA Scores by Self-Declared Investment Approach

This figure shows the distribution of TF-IDF scores for Technical Analysis (TA) words at the message level. The TA-word dictionary is obtained from [Cookson and Niessner \(2020\)](#). Each panel presents the frequency distribution of messages across levels of TA-word TF-IDF scores within each self-declared investment approach.

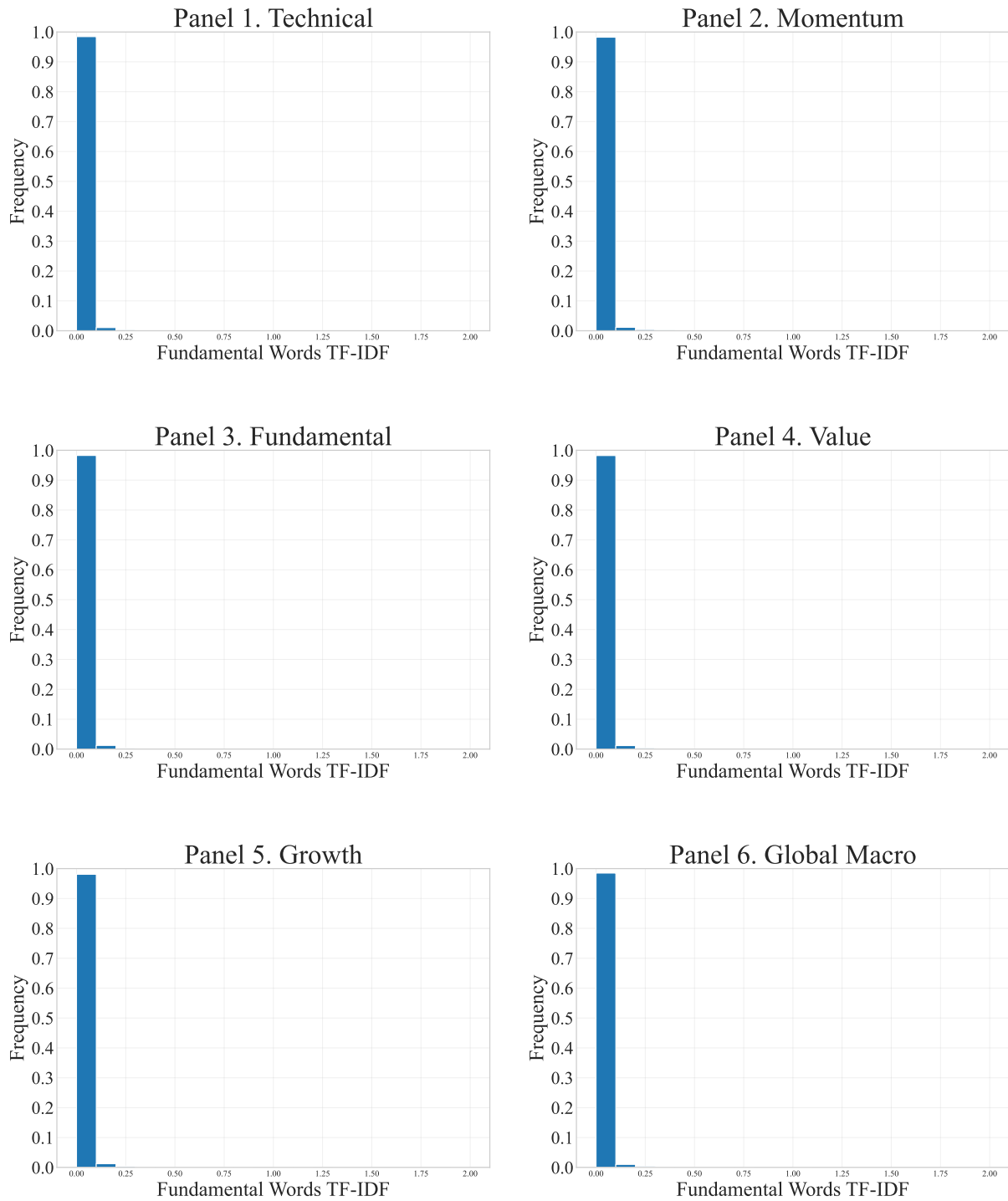


Fig. A.8. Dictionary-Based FA Scores by Self-Declared Investment Approach

This figure shows the distribution of TF-IDF scores for Fundamental Analysis (FA) words at the message level. The FA-word dictionary is obtained from [Cookson and Niessner \(2020\)](#). Each panel presents the frequency distribution of messages across levels of FA-word TF-IDF scores within each self-declared investment approach.

Table A.1: Examples of Strategy Classification

Panel A: TA Responses

No.	Message	Ticker	Score	Indicators
1	\$IOVA Biotechnology Company, Phase 2, Hammer, Support Line, Oversold, JMP Securities \$38, Q4: Institutional Bought \$77M, Sold \$13M, Speculation Trade, Entry: Above \$24	IOVA	2	Hammer, Support Line, Oversold
2	\$CVS if it can hold firmly above \$106 will signal entry at the close as well. Stops tight at \$104	CVS	2	Support Level, Stop Loss
3	\$RETA 10 wk SMA has caught up. \$300 stock btw, Livermore's finest	RETA	2	10 wk SMA
4	RT @mentholatum \$AAPL the oversold compression on AAPL will release... another \$50 up day maybe.... when???.... Someday soon// Bold call	AAPL	2	Oversold Compression
5	\$AAPL next retracement \$100.36 which is 38.2% of the move down. should be coming within next hr	AAPL	2	Fibonacci Retracement
6	\$ACOR Acorda Therapeutics (ACOR, \$8.65) was this week's top stock market loser, declining -10%. Expect a Downtrend reversal	ACOR	1	Downtrend Reversal
7	\$META Bout to break the big \$100 level then breakdown further.	META	1	Breakdown
8	\$SAIC Science Applications International Corporation (SAIC) has been systematically hitting all-time highs in the last 10 days. Science Applications International Corporation (SAIC) price climbed on Wednesday a 2.17% ending at \$103.10 and marking the n	SAIC	1	All-time Highs
9	\$PETS new retail shorts probably got in at 35 or lower, this will fly on short covering above \$38.50ish when most down over 10%	PETS	1	Short Covering
10	10:27:29 AM Makes fresh HOD \$CARA \$19.55 +12.2% ON 1,400K VOL (ISW Pre-Market Watch/Scan)	CARA	1	HOD, Volume
11	\$TSLA added more under \$890 ... well it has been while since last time I played with TSLA... I just love how their earning growing and what ELON said... I still expect volatile days but worth to start adding... GL	TSLA	0	-
12	\$MSFT Lmaooo you bears are dumb as shit. I sold all my Bitcoin to buy shares at \$275 hand over fist.	MSFT	0	-
13	\$MU I picked up some of the \$25s for a punt...Company is undervalued massively...if they deliver, this soars > 15%.	MU	0	-
14	\$ETSY at \$13.66 - Sell Stock Market Alert sent at 10:14 AM ET #stocks	ETSY	0	-

Panel B: FA Responses

No.	Message	Ticker	Score	Indicator
1	Actually nervous to see \$AAPL earnings. People expect too much and realistic is never good enough. Still fundamentally one of best stocks.	AAPL	2	earnings
2	ioDrive2 qualifications a negating effect for \$FIO revenues next quarter? BS imo. What about ioDrive which probably takes 2-3 qtrs??	FIO	2	revenues
3	\$CHK company should just put itself up for sale....assets are worth way more than the current stock price....no doubt	CHK	2	assets
4	\$NTAP not liking that discussion of non-organic rev was down 9% last yr in 1q	NTAP	2	revenues
5	Piper Jaffray details 10 Apple strengths for share price run up to \$1000 - report \$AAPL	AAPL	2	analyst-ratings
6	\$CSTR People waking up to fact \$CSTR has 2 dying businesses-- DVDs and Coincashing(anyone ever hear of debit cards and streaming)	CSTR	2	products-services
7	Rising selling margin on full-price goods is a good sign only if folks are buying more of them. Sadly not the case for Penney \$JCP	JCP	2	revenues
8	Or that it's trading at 0.5 P/B (historically trades at 1.2-2.0 P/B)? @Thinkb4trading \$AIG- does anyone realize the PE ratio is "3"?	AIG	2	assets
9	\$TSLA A lot of batteries will be needed in Florida. Quasi Republican Elon, to the rescue, selling batteries to Florida in need. Heard Generac is ready for high demand for batteries.	TSLA	1	products-services
10	\$AMC Good news they just finished Filming Honey I Shrunk the Kids 2 !!	AMC	1	products-services
11	\$TLRY my first very small position with 850 shares (bought last week) isn't printing yet... time to buy more... this is imo one of the best plays for eoy... enough catalysts in front of us... double digits and more... everybody buying options/shares here?	TLRY	1	products-services
12	Earnings whisper says \$SOFI will beat. I'm bullish on the name for growth	SOFI	1	earnings
13	in one week, may see that again, gets us to 335 level everyone is talking about \$AAPL	AAPL	0	
14	if \$AAPL dips below 435 tmrow, I'm going to jump in with some of the wkly calls - even if they are expensive - and write some more puts too	AAPL	0	

Panel C: OS Responses

No.	Message	Ticker	Score	Category
1	\$AAPL looks like it is being pegged to the \$600 quarterly option strike	AAPL	1	event-driven strategy, option strategy
2	\$AAPL BTO Jan 2013 \$650 Call @ \$64; BTC May 2012 \$750 Call @ \$3.25 & May 2012 \$775 @ \$1.9 for approx. 20% overnight gains.	AAPL	1	event-driven strategy, option strategy
3	Watching \$AAPL expiring 620 puts, tide could turn fast : \$1.25 x 1.39:	AAPL	1	event-driven strategy, option strategy
4	I think this is a pretty big negative for this stock, could test those 80 cents-\$1 lows in 2009. Really is bad news for capital plans \$GNW	GNW	1	event-driven strategy
5	Wow, any bond funds that bought the \$HCSI convert straight up in Nov must be feeling very good. Now fetching \$126 after being as low as \$84.	HGSI	1	event-driven strategy
6	\$CHK got filled on June Put Spread, got out of weekly put from Friday.. in @ .09 out today @ .23. Now long June \$17 and short June \$12	CHK	1	event-driven strategy, option strategy
7	RT @GOODGREED: \$AAPL \$610 tomorrow as shorts panic to cover...	AAPL	1	event-driven strategy
8	\$DNKN- congrats to macro investors here- been pounding the table on this one the last few month- \$33 close would be good	DNKN	1	macro
9	\$DDD as a long term options play on the Jan 2015 Contract, Sell \$40 Put, Buy \$50 Call, Sell \$55 Call, net credit apx \$3. \$\$	DDD	1	option strategy
10	\$AEO Rolled to June 17 15P/14C inverted strangle for \$0.08DB (\$1.23CR total). More time, more extrinsic value, reduce delta risk.	AEO	1	Options Trading Strategy
11	“@tunwang: \$META huge earnings on mobile, way to go. will get back to above \$50.?” Where it should belong higher. Bullish.	META	0	
12	\$META if you dumped below \$50.... good. burn with the rest pussbags	META	0	
13	\$TTWO HUGE block trades: 131355 shares traded -\$17.91 @ 3pm yesterday & 45900 shares traded -\$17.91 @ 07:50:08 today.	TTWO	0	
14	\$T What's with the recent rise of ATT? It's gone from around the \$34 range to \$36+? Someone fill me in please	T	0	
15	\$HK WTI Crude down to \$94.98 & Brent down to \$106.69. Could be impacting Halcon Resouces Corp.	HK	0	
16	\$TSLA fortunately my trigger was number hit, their system went down before .. save me \$5000 bucks. I would have stopped out regardless	TSLA	0	

Table A.2: Variable Definitions

Variable	Definition
<i>Strategy Classification and Sentiment</i>	
$Usage_{j,i,t,n}^s$	Indicator equal to one if message n , posted by user j about stock i on day t , is classified by the LLM-then-BERT procedure described in Section 2.2.1 as referencing strategy $s \in \{TA, FA, OS\}$, where TA denotes technical analysis, FA fundamental analysis, and OS other investment strategies.
$Sentiment_{i,t}^s$	Stock-day strategy-specific sentiment for $s \in \{TA, FA, OS\}$, defined as the difference between bullish and bearish strategy- s messages on stock i on day t , normalized by their sum, and set to zero on stock-days with no strategy- s messages (Cookson et al., 2024).
<i>User Attributes</i>	
Technical Investor $_j$	Indicator equal to one if user j 's self-reported investment approach is "Technical"; the omitted approaches are Momentum, Value, Growth, and Global Macro.
Fundamental Investor $_j$	Indicator equal to one if user j 's self-reported investment approach is "Fundamental".
Swing or Day Trader $_j$	Indicator equal to one if user j 's self-reported holding horizon is "Swing Trader" or "Day Trader"; the omitted horizon is Position Trader.
Long-Term Investor $_j$	Indicator equal to one if user j 's self-reported holding horizon is "Long-Term Investor".
Professional $_j$	Indicator equal to one if user j 's self-reported trading experience is "Professional"; the omitted experience level is Intermediate.
Novice $_j$	Indicator equal to one if user j 's self-reported trading experience is "Novice".
<i>Information Environment</i>	
Earnings News $_{i,t}$	Indicator equal to one if RavenPack records an earnings-related news article on stock i on day t .
Analyst News $_{i,t}$	Indicator equal to one if RavenPack records an analyst-related news article (e.g., recommendations or price targets) on stock i on day t .
Business News $_{i,t}$	Indicator equal to one if RavenPack records a business-related news article (e.g., credit-rating changes or labor issues) on stock i on day t .
<i>Past Strategy Performance and Social Feedback</i>	
High (Low) Performance $_{j,t-1}^s$	Indicator equal to one if user j 's past performance on strategy $s \in \{TA, FA, OS\}$ is at or above the 75th percentile (at or below the 25th percentile) of the pooled cross-sectional distribution. The user-strategy performance measure is the DGTW-adjusted five-day return following each s -classified message (long after positive sentiment, short after negative), averaged across the user's s -classified messages in the trailing three-month window ending five trading days before t .
High (Low) Likes $_{j,t-1}^s$	Indicator equal to one if the total number of likes received on user j 's strategy- s messages over the trailing three-month window (excluding the most recent five trading days) is at or above the 75th percentile (at or below the 25th percentile) of the pooled cross-sectional distribution.
<i>Stock-Day Retail-Trading and Attention Variables</i>	
Attention $_{i,t}$	Number of StockTwits messages on stock i on day t scaled by the total number of messages posted that day (Cookson et al., 2024).
FracProMsg $_{i,t}$	Fraction of messages on stock i on day t posted by users with a self-reported "Professional" trading experience.
$OIB_{i,t}^{BJZZ}$	Net retail order imbalance on stock i on day t , defined as marketable retail buy dollar volume minus marketable retail sell dollar volume, scaled by their sum, with retail orders identified following Boehmer et al. (2021).
$OIB_{i,t}^{BHJOS}$	Net retail order imbalance constructed analogously to $OIB_{i,t}^{BJZZ}$, with retail orders identified following Barber et al. (2023a).
RH_Herd $_{i,t}$	Indicator equal to one if stock i is among the top ten stocks on day t ranked by the daily percentage increase in the number of Robinhood users holding the stock, conditional on at least 100 users holding the stock at the end of day $t - 1$ (Barber et al., 2022).
Squeeze $_{i,t}$	Indicator equal to one if stock i experiences a short-squeeze event on day t , defined as a stock-day in which the one-day stock return exceeds 30% and the daily percentage change in short loan value decreases by more than 20% (Schultz, 2024).
<i>Firm Characteristics</i>	
MAX $_{i,t}$	Maximum daily return on stock i in the preceding month (Bali et al., 2011).
Abnormal Turnover $_{i,t}$	$\log(\text{Turnover}_{i,t}) - \log(\frac{1}{4} \sum_{h=1}^4 \text{Turnover}_{i,t-h})$.
Abnormal News Volume $_{i,t}$	$\log(1 + \#News_{i,t}) - \log(1 + \frac{1}{4} \sum_{h=1}^4 \#News_{i,t-h})$, where $\#News_{i,t}$ is the count of RavenPack news articles on stock i on day t .
Market Capitalization $_{i,t}$	Stock price times shares outstanding.
Book-to-Market $_{i,t}$	Ratio of book value of equity to market value of equity.
Asset Growth $_{i,t}$	Annual growth rate of total assets.
Gross Profit-to-Assets $_{i,t}$	Ratio of gross profit to total assets.
Analyst Coverage $_{i,t}$	Number of IBES equity analysts covering stock i .
Institutional Ownership $_{i,t}$	Fraction of shares outstanding held by 13F institutional investors.
<i>Benchmark Classifiers (Appendix A.1)</i>	
Technical $_{j,i,t,n}^{TF-IDF}$	TF-IDF score of technical-analysis keywords in message n , computed using the keyword list of Cookson and Niessner (2020).
Fundamental $_{j,i,t,n}^{TF-IDF}$	TF-IDF score of fundamental-analysis keywords in message n , computed using the keyword list of Cookson and Niessner (2020).

Table A.3: **Additional Summary Statistics**

This table reports summary statistics for the two strategy-choice samples used in the analysis. The sample period is January 2010 to June 2023. Panel A reports statistics for the message-level sample, restricted to StockTwits messages from users with valid self-reported profile information. $Usage^{TA}$ ($Usage^{FA}$) indicates whether a message references Technical Analysis (TA) or Fundamental Analysis (FA). *Technical Investor*, *Fundamental Investor*, *Swing or Day Trader*, *Long-Term Investor*, *Professional*, and *Novice* are user-type indicators from self-reported profiles. *Earnings News*, *Analyst News*, and *Business News* are stock-day news indicators from RavenPack. Panel B reports statistics for the active-user subsample, where we require the availability of TA and FA messages in the prior quarter. *TA (FA) Performance*, in percent, is the user's prior-quarter average DGTW-adjusted return on TA (FA) messages; *High/Low Performance* indicators flag users above the 75th or below the 25th percentile of the cross-sectional distribution. *TA (FA) Likes* is the total number of likes received on TA (FA) messages; *High/Low Likes* indicators are constructed analogously. Variable definitions are provided in Appendix Table A.2.

Panel A: Message-Level Sample: Self-Reported User Information				
	N	Mean	StdDev	Median
$Usage^{TA}$	21,847,364	0.14	0.34	0
$Usage^{FA}$	21,847,364	0.17	0.37	0
Technical Investor	21,847,364	0.36	0.48	0
Fundamental Investor	21,847,364	0.14	0.34	0
Swing or Day Trader	21,847,364	0.59	0.49	1
Long-Term Investor	21,847,364	0.22	0.42	0
Professional	21,847,364	0.32	0.46	0
Novice	21,847,364	0.14	0.35	0
Earnings News	21,847,364	0.17	0.37	0
Analyst News	21,847,364	0.19	0.39	0
Business News	21,847,364	0.46	0.50	0

Panel B: Active-User Sample for Strategy Malleability Analysis				
	N	Mean	StdDev	Median
Usage ^{TA}	4,717,310	0.20	0.40	0
Usage ^{FA}	4,717,310	0.21	0.41	0
TA Performance (%)	4,717,310	-0.37	9.33	-0.34
FA Performance (%)	4,717,310	-0.08	9.73	-0.30
High TA Performance	4,717,310	0.25	0.43	0
Low TA Performance	4,717,310	0.25	0.43	0
High FA Performance	4,717,310	0.25	0.43	0
Low FA Performance	4,717,310	0.25	0.43	0
TA Likes	4,717,310	254	928	44
FA Likes	4,717,310	203	502	56
High TA Likes	4,717,310	0.25	0.43	0
Low TA Likes	4,717,310	0.26	0.44	0
High FA Likes	4,717,310	0.25	0.43	0
Low FA Likes	4,717,310	0.25	0.44	0

Table A.4: **Strategy Sentiment and Herding Reversals**

This table reports stock-day predictive regressions of future returns on intense buying episodes by Robinhood investors and the interaction of those episodes with StockTwits strategy-specific sentiments, corresponding to the specification in Eq. (8). The dependent variable $R_{i,h}$ is the cumulative stock return over horizon h : $h = t + 1$ in columns (1)–(2) and $t + 2$ to $t + 5$ in column (3). Column (1) includes only $RH_Herd_{i,t}$; columns (2)–(3) additionally include $Sentiment_{i,t}^{TA}$, $Sentiment_{i,t}^{FA}$, and their interactions with $RH_Herd_{i,t}$. $RH_Herd_{i,t}$ is an indicator equal to one if stock i is among the top ten stocks ranked by the daily percentage increase in Robinhood users holding the stock, conditional on at least 100 users holding the stock at the end of the previous day (Barber et al., 2022); the holdings data, from RobinTrack, span May 2018 to August 2020. Sentiment variables, *Attention*, and the firm-level controls $X_{i,t}$ are defined as in Table 5. All specifications include date fixed effects. The dependent variable is multiplied by 10,000 in the regressions, so each coefficient measures the change in cumulative return, in basis points, associated with a unit change in the regressor. Standard errors are clustered by stock and trading day, with t -statistics in brackets. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Return _{$i,t+1$}		Return _{$i,t+2 \rightarrow t+5$}
	(1)	(2)	(3)
$Sentiment_{i,t}^{TA} \times RH_Herd_{i,t}$		-159.41*** [-3.20]	-54.96 [-0.86]
$Sentiment_{i,t}^{FA} \times RH_Herd_{i,t}$		-94.70 [-1.49]	-45.30 [-0.68]
$Sentiment_{i,t}^{TA}$		2.75* [1.67]	5.16 [1.41]
$Sentiment_{i,t}^{FA}$		3.65** [2.24]	6.12* [1.92]
$RH_Herd_{i,t}$	-75.48** [-2.10]	28.38 [0.70]	-77.38 [-1.41]
$Attention_{i,t}$	0.18 [0.04]	0.36 [0.08]	10.32 [0.99]
Stock Characteristics	Yes	Yes	Yes
Lagged Returns	Yes	Yes	Yes
Date FEs	Yes	Yes	Yes
N	554,799	554,799	554,569
Adj. R ² (%)	9.74	9.75	10.36

Table A.5: **Strategy Sentiment and Squeeze Reversals**

This table reports stock-day predictive regressions of future returns on short-squeeze episodes and the interaction of those episodes with StockTwits strategy-specific sentiments. The dependent variable $R_{i,h}$ is the cumulative stock return over horizon h : $h = t + 1$ in columns (1)–(2) and $t + 2$ to $t + 5$ in column (3). Column (1) includes only $Squeeze_{i,t}$; columns (2)–(3) additionally include $Sentiment_{i,t}^{TA}$, $Sentiment_{i,t}^{FA}$, and their interactions with $Squeeze_{i,t}$. $Squeeze_{i,t}$ is an indicator equal to one if stock i experiences a short-squeeze event on day t , defined as a stock-day in which the one-day stock return exceeds 30% and the daily percentage change in short loan value decreases more than 20%. Sentiment variables, *Attention*, and the firm-level controls $X_{i,t}$ are defined as in Table 5. All specifications include date fixed effects. The dependent variable is multiplied by 10,000 in the regressions, so each coefficient measures the change in cumulative return, in basis points, associated with a unit change in the regressor. Standard errors are clustered by stock and trading day, with t -statistics in brackets. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Return _{$i,t+1$}		Return _{$i,t+2 \rightarrow t+5$}
	(1)	(2)	(3)
$Sentiment_{i,t}^{TA} \times Squeeze_{i,t}$		-211.20*	-263.83**
		[-1.75]	[-2.29]
$Sentiment_{i,t}^{FA} \times Squeeze_{i,t}$		10.70	173.46
		[0.08]	[1.34]
$Sentiment_{i,t}^{TA}$		-1.61**	-5.29***
		[-2.20]	[-3.18]
$Sentiment_{i,t}^{FA}$		1.57**	2.17
		[2.55]	[1.59]
$Squeeze_{i,t}$	-601.56***	-505.26***	-457.85***
	[-8.53]	[-5.42]	[-4.41]
$Attention_{i,t}$	-4.79*	-4.77*	-9.75
	[-1.76]	[-1.76]	[-1.64]
Stock Characteristics	Yes	Yes	Yes
Lagged Returns	Yes	Yes	Yes
Date FEs	Yes	Yes	Yes
N	2,958,286	2,958,286	2,956,927
Adj. R ² (%)	8.85	8.85	10.69

Table A.6: Subperiods

This table reports results from stock-day predictive regressions of future returns on StockTwits strategy-based sentiment, estimated separately by return horizon and subperiod. The dependent variable is the cumulative stock return over horizon h , with $h = t + 1$ in columns (1)–(2), $t + 2$ to $t + 5$ in columns (3)–(4), and $t + 6$ to $t + 10$ in columns (5)–(6). Within each horizon, the regression is estimated separately for two subperiods: 2010–2017 and 2018–2023. The control variables are the same as in Table 5. All specifications include trading day fixed effects. The dependent variable is multiplied by 10,000 in the regressions, so each coefficient measures the change in cumulative return, in basis points, associated with a unit change in the regressor. Standard errors are clustered by trading day, with corresponding t -statistics in brackets. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Return _{$i,t+1$}		Return _{$i,t+2 \rightarrow t+5$}		Return _{$i,t+6 \rightarrow t+10$}	
	(1)	(2)	(3)	(4)	(5)	(6)
Sentiment _{i,t} ^{TA}	0.07	-3.04***	-3.38**	-6.84***	-6.89***	-5.00**
	[0.10]	[-2.78]	[-2.33]	[-3.08]	[-4.19]	[-2.06]
Sentiment _{i,t} ^{FA}	0.70	2.17**	1.86	2.59	-0.88	-0.42
	[0.95]	[2.54]	[1.27]	[1.54]	[-0.59]	[-0.22]
Attention _{i,t}	-3.63***	-13.11***	-7.80***	-20.97***	-6.14***	-17.31***
	[-4.36]	[-3.51]	[-6.27]	[-3.83]	[-6.09]	[-3.25]
Stock Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Lagged Returns	Yes	Yes	Yes	Yes	Yes	Yes
Date FEs	Yes	Yes	Yes	Yes	Yes	Yes
Subperiod	2010–2017	2018–2023	2010–2017	2018–2023	2010–2017	2018–2023
N	1,121,217	1,853,087	1,120,443	1,851,894	1,119,629	1,850,584
Adj. R ² (%)	7.39	9.20	8.83	11.28	8.28	11.50

Table A.7: **Alternative Sentiment Measure**

This table reports results from stock-day predictive regressions of future returns on strategy-specific sentiment, where sentiment is constructed using the rule-based algorithm of [Cookson and Niessner \(2020\)](#) rather than the BERT-based classification used in the main analysis. The dependent variable is the cumulative stock return over horizon h , with $h = t + 1$ in column (1), $t + 2$ to $t + 5$ in column (2), $t + 6$ to $t + 10$ in column (3), and $t + 11$ to $t + 15$ in column (4). The control variables are the same as in Table 5. All specifications include trading day fixed effects. The dependent variable is multiplied by 10,000 in the regressions, so each coefficient measures the change in cumulative return, in basis points, associated with a unit change in the regressor. Standard errors are clustered by trading day, with corresponding t -statistics in brackets. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Return _{$i,t+1$}	Return _{$i,t+2 \rightarrow t+5$}	Return _{$i,t+6 \rightarrow t+10$}	Return _{$i,t+11 \rightarrow t+15$}
	(1)	(2)	(3)	(4)
Sentiment _{i,t} ^{TA}	-2.56*** [-3.52]	-5.87*** [-3.81]	-4.45*** [-2.67]	-2.91* [-1.79]
Sentiment _{i,t} ^{FA}	1.38** [2.20]	-1.39 [-1.09]	0.20 [0.14]	0.94 [0.68]
Attention _{i,t}	-5.57*** [-5.46]	-10.31*** [-6.81]	-8.32*** [-6.07]	-6.52*** [-5.37]
Stock Characteristics	Yes	Yes	Yes	Yes
Lagged Returns	Yes	Yes	Yes	Yes
Date FEs	Yes	Yes	Yes	Yes
N	2,974,304	2,972,337	2,970,213	2,968,174
Adj. R ² (%)	8.76	10.68	10.67	11.14

Table A.8: Fama-MacBeth Regressions

This table reports Fama-MacBeth estimates of stock-day predictive regressions of future returns on StockTwits strategy-based sentiment. The dependent variable is the cumulative stock return over horizon h , with $h = t + 1$ in column (1), $t + 2$ to $t + 5$ in column (2), $t + 6$ to $t + 10$ in column (3), and $t + 11$ to $t + 15$ in column (4). The control variables are the same as in Table 5. The dependent variable is multiplied by 10,000 in the regressions, so each coefficient measures the change in cumulative return, in basis points, associated with a unit change in the regressor. Standard errors are Newey-West with one lag, with corresponding t -statistics in brackets. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Return _{$i,t+1$}	Return _{$i,t+2 \rightarrow t+5$}	Return _{$i,t+6 \rightarrow t+10$}	Return _{$i,t+11 \rightarrow t+15$}
	(1)	(2)	(3)	(4)
Sentiment _{i,t} ^{TA}	-0.96 [-1.55]	-4.71*** [-3.71]	-5.68*** [-3.99]	-1.77 [-1.33]
Sentiment _{i,t} ^{FA}	1.31** [2.20]	1.66 [1.42]	-1.37 [-1.08]	-1.05 [-0.85]
Attention _{i,t}	-9.36*** [-5.37]	-15.83*** [-4.59]	-12.35*** [-3.63]	-10.36*** [-3.43]
Stock Characteristics	Yes	Yes	Yes	Yes
Lagged Returns	Yes	Yes	Yes	Yes
N	2,974,304	2,972,337	2,970,213	2,968,174
Adj. R ² (%)	7.46	7.37	6.71	6.74

Table A.9: **DGTW-Adjusted Returns**

This table reports results from stock-day predictive regressions of future DGTW-adjusted abnormal returns (Daniel et al., 1997) on StockTwits strategy-based sentiment. The dependent variable is the cumulative DGTW-adjusted return over horizon h , with $h = t + 1$ in column (1), $t + 2$ to $t + 5$ in column (2), $t + 6$ to $t + 10$ in column (3), and $t + 11$ to $t + 15$ in column (4). The control variables are the same as in Table 5, except that the lagged return controls are replaced by lagged DGTW-adjusted returns. All specifications include trading day fixed effects. The dependent variable is multiplied by 10,000 in the regressions, so each coefficient measures the change in cumulative DGTW-adjusted return, in basis points, associated with a unit change in the regressor. Standard errors are clustered by trading day, with corresponding t -statistics in brackets. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	DGTW _{$i,t+1$}	DGTW _{$i,t+2 \rightarrow t+5$}	DGTW _{$i,t+6 \rightarrow t+10$}	DGTW _{$i,t+11 \rightarrow t+15$}
	(1)	(2)	(3)	(4)
Sentiment _{i,t} ^{TA}	-1.96*** [-3.01]	-4.70*** [-3.62]	-6.17*** [-4.42]	-2.72** [-1.97]
Sentiment _{i,t} ^{FA}	1.48*** [2.59]	2.49** [2.25]	-0.12 [-0.09]	-0.40 [-0.33]
Attention _{i,t}	-4.19*** [-4.77]	-5.01*** [-3.88]	-4.70*** [-4.21]	-2.32** [-2.48]
Stock Characteristics	Yes	Yes	Yes	Yes
Lagged Returns	Yes	Yes	Yes	Yes
Date FEs	Yes	Yes	Yes	Yes
N	2,665,356	2,663,028	2,660,293	2,657,567
Adj. R ² (%)	0.50	0.79	0.77	0.79

Table A.10: **Investor Sophistication**

This table examines how investor sophistication influences the return predictability of Stock-Twits strategy-based sentiment. Extending the predictive regression in Table 5, we augment the specification with $FracProMsg_{i,t}$, the fraction of messages on stock i on day t posted by self-declared professional investors, and its interactions with TA and FA sentiment. The dependent variable is the cumulative stock return over horizon h , with $h = t + 1$ in column (1), $t + 2$ to $t + 5$ in column (2), $t + 6$ to $t + 10$ in column (3), and $t + 11$ to $t + 15$ in column (4). The control variables are the same as in Table 5. All specifications include trading day fixed effects. The dependent variable is multiplied by 10,000 in the regressions, so each coefficient measures the change in cumulative return, in basis points, associated with a unit change in the regressor. Standard errors are clustered by trading day, with corresponding t -statistics in brackets. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Return _{$i,t+1$}	Return _{$i,t+2 \rightarrow t+5$}	Return _{$i,t+6 \rightarrow t+10$}	Return _{$i,t+11 \rightarrow t+15$}
	(1)	(2)	(3)	(4)
Sentiment _{i,t} ^{TA} $\times$ FracProMsg _{i,t}	3.03** [2.07]	9.01*** [2.97]	13.48*** [3.88]	6.80** [2.05]
Sentiment _{i,t} ^{FA} $\times$ FracProMsg _{i,t}	2.16 [1.42]	1.43 [0.45]	-2.27 [-0.65]	-0.05 [-0.02]
Sentiment _{i,t} ^{TA}	-2.58*** [-2.76]	-7.68*** [-3.94]	-9.56*** [-4.45]	-3.70* [-1.74]
Sentiment _{i,t} ^{FA}	1.09 [1.38]	2.03 [1.30]	0.24 [0.14]	-0.21 [-0.12]
FracProMsg _{i,t}	2.14** [2.33]	4.82** [2.41]	2.91 [1.29]	2.68 [1.23]
Attention _{i,t}	-5.52*** [-5.41]	-10.26*** [-6.80]	-8.20*** [-5.99]	-6.44*** [-5.30]
Stock Characteristics	Yes	Yes	Yes	Yes
Lagged Returns	Yes	Yes	Yes	Yes
Date FEs	Yes	Yes	Yes	Yes
N	2,974,304	2,972,337	2,970,213	2,968,174
Adj. R ² (%)	8.76	10.68	10.67	11.14

Table A.11: **Other Strategy Choice**

This table extends the strategy-choice analyses of Tables 2, 3, and 4 to messages classified as referencing other strategies (OS). The dependent variable is $Usage_{j,i,t,n}^{OS}$, an indicator equal to one if message n , posted by user j about stock i on day t , is classified as referencing an investment strategy distinct from TA and FA, and zero otherwise. Column (1) regresses $Usage^{OS}$ on self-reported user attributes, mirroring Eq. (1); column (2) replaces user attributes with public-news indicators and adds user fixed effects, mirroring Eq. (2); column (3) augments the specification with past TA, FA, and OS performance, mirroring Eq. (3); and column (4) further adds past TA, FA, and OS likes, mirroring Eq. (4). All right-hand-side variables are defined as in the corresponding main-text tables. The dependent variable is multiplied by 100 in the regressions, so each coefficient measures the percentage-point change in $Usage^{OS}$ associated with a unit change in the regressor. Standard errors are clustered by user, stock, and trading day, with t -statistics in brackets. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	$Usage_{j,i,t,n}^{OS}$			
	(1)	(2)	(3)	(4)
Technical Investor	0.87*** [3.59]			
Fundamental Investor	-0.56** [-1.99]			
Swing or Day Trader	1.17*** [5.45]			
Long-Term Investor	-1.52*** [-6.63]			
Professional	1.66*** [6.31]			
Novice	-1.30*** [-8.68]			
Earnings News $_{i,t}$		-0.74*** [-11.26]	-1.16*** [-7.80]	-1.17*** [-7.83]
Analyst News $_{i,t}$		-0.32*** [-6.47]	-0.61*** [-5.73]	-0.61*** [-5.73]
Business News $_{i,t}$		-0.33*** [-7.65]	-0.80*** [-7.36]	-0.80*** [-7.37]
High TA Performance $_{j,t-1}$			-0.02 [-0.22]	-0.03 [-0.23]
Low TA Performance $_{j,t-1}$			-0.15 [-0.77]	-0.16 [-0.82]
High FA Performance $_{j,t-1}$			0.05 [0.47]	0.02 [0.20]
Low FA Performance $_{j,t-1}$			0.23** [2.05]	0.19* [1.79]
High OS Performance $_{j,t-1}$			-0.09 [-0.67]	-0.10 [-0.80]
Low OS Performance $_{j,t-1}$			-0.01 [-0.08]	-0.02 [-0.16]
High TA Likes $_{j,t-1}$				-0.31 [-1.42]
Low TA Likes $_{j,t-1}$				0.44*** [2.96]
High FA Likes $_{j,t-1}$				-0.41** [-2.18]
Low FA Likes $_{j,t-1}$				0.51** [2.55]
High OS Likes $_{j,t-1}$				0.09 [0.34]
Low OS Likes $_{j,t-1}$				-0.17 [-1.09]
Past Strategy Usage	No	No	Yes	Yes
User FEs	No	Yes	Yes	Yes
Stock FEs	Yes	Yes	Yes	Yes
Date FEs	Yes	Yes	Yes	Yes
N	21,847,221	21,829,799	3,570,610	3,570,610
Adj. R ² (%)	1.24	5.95	3.26	3.27

Table A.12: **Other-Strategy Sentiment: Return Predictability and Retail Order Imbalance**

This table extends the asset-pricing analyses to OS sentiment, $Sentiment_{i,t}^{OS}$, and to NS sentiment, $Sentiment_{i,t}^{NS}$, constructed from messages that do not reference any of TA, FA, or OS, controlling for TA and FA sentiment throughout. Column (1) re-estimates the next-day predictive regression in Eq. (5) with TA, FA, OS, and NS sentiment entered jointly; the dependent variable is the next-day stock return $Return_{i,t+1}$. Columns (2) and (3) re-estimate the contemporaneous OIB regression in Eq. (10) with the same four sentiment series entered jointly; column (2) uses $OIB_{i,t}^{BJZZ}$ and column (3) uses $OIB_{i,t}^{BHJOS}$. To align the sentiment measures with intraday retail orders, strategy-specific sentiment in columns (2) and (3) is constructed using only messages posted during regular trading hours (9:30–16:00 ET). The control variables are the same as in Table 5. All specifications include trading day fixed effects. The dependent variable is multiplied by 10,000 in column (1) and by 100 in columns (2) and (3), so each coefficient measures the change in next-day return (in basis points) or in net retail buying (in percentage points) associated with a unit change in the regressor. Standard errors are clustered by trading day in column (1) and by stock and trading day in columns (2) and (3), with t -statistics in brackets. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Return _{<i>i,t</i>+1}	OIB _{<i>i,t</i>} ^{BJZZ}	OIB _{<i>i,t</i>} ^{BHJOS}
	(1)	(2)	(3)
Sentiment _{<i>i,t</i>} ^{OS}	-2.61*** [-3.73]	0.72*** [19.06]	0.79*** [22.89]
Sentiment _{<i>i,t</i>} ^{NS}	-0.16 [-0.30]	0.49*** [13.57]	0.63*** [18.20]
Sentiment _{<i>i,t</i>} ^{TA}	-1.55** [-2.25]	0.73*** [17.48]	0.98*** [24.97]
Sentiment _{<i>i,t</i>} ^{FA}	1.73*** [2.88]	0.57*** [15.02]	0.61*** [16.33]
Attention _{<i>i,t</i>}	-5.57*** [-5.47]	0.39*** [2.75]	0.48*** [3.38]
Stock Characteristics	Yes	Yes	Yes
Lagged Returns	Yes	Yes	Yes
Date FEs	Yes	Yes	Yes
N	2,974,304	2,974,934	2,974,934
Adj. R ² (%)	8.76	0.74	1.06

Table A.13: **Strategy Choice and Social Feedback (Showing Controls)**

This table reports message-level panel regressions that examine how a user's strategy choice responds to the social feedback (likes) received on recent strategy-classified messages, corresponding to the specification in Equation (4). The dependent variable $Usage_{j,i,t,n}^s$ equals one if message n , posted by user j about stock i on day t , is classified as referencing strategy $s \in \{TA, FA\}$, with $s = TA$ in columns (1)–(2) and $s = FA$ in columns (3)–(4). *High TA Likes* $_{j,t-1}$ (*Low TA Likes* $_{j,t-1}$) equals one if the total number of likes user j received on TA messages posted in the prior three months (excluding the most recent five trading days) is at or above the 75th percentile (at or below the 25th percentile) of the pooled cross-sectional distribution of TA-message likes, and zero otherwise. *High FA Likes* $_{j,t-1}$ and *Low FA Likes* $_{j,t-1}$ are defined analogously for FA messages. All specifications control for public-news indicators (*Earnings News*, *Analyst News*, *Business News*) and the user's recent intensity of TA and FA usage over the prior three months. Columns (2) and (4) additionally control for *High/Low TA Performance* $_{j,t-1}$ and *High/Low FA Performance* $_{j,t-1}$ (defined as in Table 3). All specifications include user, stock, and date fixed effects. The sample is restricted to users who posted at least one strategy-related (TA or FA) message in the prior three months. The dependent variable is multiplied by 100 in the regressions, so each coefficient measures the percentage-point change in *Usage* associated with a unit change in the regressor. Standard errors are triple-clustered by date, stock, and user, with t -statistics in brackets. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Usage $_{j,i,t,n}^{TA}$		Usage $_{j,i,t,n}^{FA}$	
	(1)	(2)	(3)	(4)
High TA Likes	0.85*** [3.04]	0.84*** [3.02]	-0.87*** [-2.96]	-0.85*** [-2.88]
Low TA Likes	-0.83*** [-4.01]	-0.82*** [-4.05]	1.13*** [5.30]	1.11*** [5.09]
High FA Likes	-1.43*** [-6.16]	-1.42*** [-6.13]	1.33*** [3.54]	1.30*** [3.49]
Low FA Likes	1.70*** [8.27]	1.69*** [8.24]	-1.35*** [-5.95]	-1.33*** [-5.74]
High TA Performance		-0.19 [-1.23]		0.20 [1.42]
Low TA Performance		-0.40** [-2.30]		0.80*** [2.79]
High FA Performance		0.05 [0.36]		-0.32* [-1.91]
Low FA Performance		0.30** [1.97]		-0.58*** [-3.05]
Earnings News	-3.17*** [-11.94]	-3.17*** [-11.95]	5.41*** [12.86]	5.41*** [12.87]
Analyst News	-2.37*** [-13.37]	-2.37*** [-13.36]	2.05*** [6.66]	2.05*** [6.68]
Business News	-3.06*** [-15.03]	-3.06*** [-15.03]	2.20*** [7.88]	2.20*** [7.88]
Past 3-Month TA Usage	0.01*** [5.50]	0.01*** [5.39]	-0.00*** [-2.64]	-0.00*** [-2.63]
Past 3-Month FA Usage	-0.00*** [-3.93]	-0.00*** [-3.93]	0.01*** [2.66]	0.01*** [2.67]
User FEs	Yes	Yes	Yes	Yes
Stock FEs	Yes	Yes	Yes	Yes
Date FEs	Yes	Yes	Yes	Yes
N	4,716,661	4,716,661	4,716,661	4,716,661
Adj. R ² (%)	10.01	10.01	10.32	10.32

Table A.14: **Strategy Sentiment and Next-Day Returns (Showing Controls)**

This table reports stock-day predictive regressions of next-day returns on strategy-specific sentiment over January 2010 to June 2023, corresponding to the specification in Eq. (5). The dependent variable $Return_{i,t+1}$ is the next-day return on stock i . $Sentiment_{i,t}^s$ is the stock-day sentiment for strategy $s \in \{TA, FA\}$, computed as the difference between the bullish and bearish message counts in category s , scaled by the total number of messages in that category. Columns (1) and (2) include TA and FA sentiment separately; column (3) includes both jointly. $Attention_{i,t}$ is the share of total StockTwits messages posted on day t that reference stock i . The control vector $X_{i,t}$ comprises $Log(\text{Market Cap})$, $Log(\text{Book-to-Market})$, $Asset\ Growth$, $Gross\ Profit\text{-to-Assets}$, $Log(1+\text{Analyst Coverage})$, $Log(\text{Institutional Ownership})$, MAX , $Abnormal\ Turnover$, $Abnormal\ News\ Volume$, and the five most recent daily return lags. All specifications include date fixed effects. The dependent variable is multiplied by 10,000 in the regressions, so each coefficient measures the change in next-day return, in basis points, associated with a unit change in the regressor. Standard errors are clustered by trading day, with t -statistics in brackets. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Return _{<i>i,t+1</i>}		
	(1)	(2)	(3)
Sentiment _{<i>i,t</i>} ^{TA}	-1.61** [-2.19]		-1.81** [-2.52]
Sentiment _{<i>i,t</i>} ^{FA}		1.37** [2.17]	1.57** [2.56]
Attention _{<i>i,t</i>}	-5.58*** [-5.47]	-5.61*** [-5.50]	-5.60*** [-5.49]
Log(Market Cap _{<i>i,t</i>})	-1.77*** [-3.22]	-1.74*** [-3.16]	-1.76*** [-3.19]
Log(Book-to-Market _{<i>i,t</i>})	0.83 [1.05]	0.87 [1.10]	0.86 [1.09]
Asset Growth _{<i>i,t</i>}	-0.70 [-0.80]	-0.75 [-0.86]	-0.72 [-0.83]
Gross Profit-to-Assets _{<i>i,t</i>}	26.82*** [3.69]	27.36*** [3.76]	27.22*** [3.75]
Log(1+Analyst Coverage _{<i>i,t</i>})	2.73*** [2.66]	2.71*** [2.65]	2.74*** [2.67]
Log(Institutional Ownership _{<i>i,t</i>})	1.03 [1.05]	1.10 [1.11]	1.04 [1.06]
MAX _{<i>i,t</i>}	-24.72** [-2.33]	-25.48** [-2.40]	-24.84** [-2.35]
Abnormal Turnover _{<i>i,t</i>}	-2.01* [-1.91]	-2.06* [-1.95]	-2.02* [-1.92]
Abnormal News Volume _{<i>i,t</i>}	0.42 [1.09]	0.42 [1.09]	0.41 [1.06]
Return _{<i>i,t-1</i>}	-81.81*** [-3.26]	-82.85*** [-3.30]	-82.06*** [-3.27]
Return _{<i>i,t-2</i>}	-81.08*** [-3.91]	-81.46*** [-3.93]	-81.22*** [-3.92]
Return _{<i>i,t-3</i>}	-54.39** [-2.55]	-54.61** [-2.56]	-54.50** [-2.56]
Return _{<i>i,t-4</i>}	-82.49*** [-4.65]	-82.65*** [-4.66]	-82.56*** [-4.66]
Return _{<i>i,t-5</i>}	-44.88** [-2.41]	-45.02** [-2.41]	-44.94** [-2.41]
Date FEs	Yes	Yes	Yes
N	2,974,304	2,974,304	2,974,304
Adj. R ² (%)	8.76	8.76	8.76